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Interpreting field measurements of juvenile growth and survival rates with population growth isoclines

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Complete List of Authors:	Barrus, Nathan; Florida International University, Institute of Environment and Department of Biological Sciences; Florida Atlantic University Cook, Mark; South Florida Water Management District, Applied Science Bureau Dorn, Nathan; Florida International University, Institute of Environment and Department of Biological Sciences; Florida Atlantic University, Department of Biological Sciences
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seasonal environmental conditions (i.e., water depth and temperature). Daily juvenile survival rates were lower in the cooler dry season than in the warmer wet (rainy) season. Juvenile growth was faster in the warmer wet season. Parameter combinations of juvenile growth and survival in the dry season predicted declining populations ($\lambda < 1$), while rates from wet season predicted populations at replacement ($\lambda = 1$) or increasing. When parameters were combined for the full annual recruitment window, populations were projected to decline in both wetlands. The qualitative predictions were robust to variation in hydrologic conditions affecting reproductive rates, but with better hydrologic conditions one lambda was near one (i.e., at replacement). Our demographic isoclines gave population-dynamic context to field measured demographic rates, identified important temporal variation in survival and growth for the population and generated new hypotheses for future investigation and management. We encourage others to consider developing demographic isocline to interpret variation of stage-sensitive demographic rates across spatiotemporal environmental conditions.

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Interpreting field measurements of juvenile growth and survival rates with population growth isoclines

Nathan T. Barrus[1,2,3]; ORCID: 0000-0001-7503-3120

Mark I. Cook[4]; ORCID: 0009-0002-4206-7548

and Nathan J. Dorn[2,3]; ORCID: 0000-0001-5516-0253

[1] Corresponding Author: Nathan T. Barrus, nbarrus1@gmail.com

[2] Current affiliation: Institute of Environment and Dept. of Biological Sciences, Florida International University, Miami, FL, USA

[3] Department of Biological Sciences, Florida Atlantic University, Boca Raton, FL, USA

[4] Applied Sciences Bureau, South Florida Water Management District. West Palm Beach, FL, USA Open Research Statement

Open Research Statement

Upon acceptance, all data and code associated with the manuscript will be provided via Zenodo.

Key words

Consumptive effects, size-dependent mortality, demographic rates, interaction strength, predator-prey, top-down vs bottom up.

Abstract:

Juvenile survival and growth rates are commonly studied demographic rates with consequences for population growth. For species that can grow to achieve a size refuge from juvenile predators, the time spent at smaller vulnerable sizes is expected to affect population dynamics. But the interactive effects of juvenile growth and survival on population growth are rarely illustrated theoretically, and most studies of these concepts have been in experimental settings. The interactive effects of the two rates have applications to field studies of recruitment variation for a diversity of species that could be assessed with demographic models and isoclines. We conceptually illustrate the potential of using demographic isoclines for marine, terrestrial and freshwater examples in the literature, and then demonstrate the use of a demographic isocline in a case study for an annual freshwater gastropod (Florida Apple Snail, *Pomacea paludosa*). Using a published size-indexed demographic model, we constructed a zero-population growth isocline for theoretical combinations of juvenile growth and survival rates. We then quantified daily juvenile survival and growth in two wetlands twice during the recruitment period, incorporating variable predator assemblages and seasonal environmental conditions (i.e., water depth and temperature). Daily juvenile survival rates were lower in the cooler dry season than in the warmer wet (rainy) season. Juvenile growth was faster in the warmer wet season. Parameter combinations of juvenile growth and survival in the dry season predicted declining populations ($\lambda < 1$), while rates from wet season predicted populations at replacement ($\lambda = 1$) or increasing. When parameters were combined for the full annual recruitment window, populations were projected to decline in both wetlands. The qualitative predictions were robust to variation in hydrologic conditions affecting reproductive rates, but with better hydrologic conditions one lambda was near one (i.e., at replacement). Our demographic isoclines gave population-dynamic

context to field measured demographic rates, identified important temporal variation in survival and growth for the population and generated new hypotheses for future investigation and management. We encourage others to consider developing demographic isocline to interpret variation of stage-sensitive demographic rates across spatiotemporal environmental conditions.

Introduction

Population growth dynamics for many species are influenced by the interaction between individual growth rates and stage- or size-specific mortality rates (Werner and Gilliam 1984, De Roos et al. 2003, Craig et al. 2006). In a review paper 40 years ago, Werner and Gilliam (1984) wrote that size-indexed demographic models are important because a) size is a key feature affecting vital rates, and b) growth rates determine the relationship between size and age. Growth rates and size-dependent survival rates interact to determine the overall number of deaths for a cohort through at least two mechanisms: 1) higher growth rates during favorable seasons can decrease individual mortality rates through a subsequent stressful season because larger individuals tend to survive stressful conditions better than smaller individuals (Carlson et al. 2008, Viñals-Domingo et al. 2020); and, 2) growth can be a type of defense against stage-specific consumers because growth determines the amount of time an individual spends in vulnerable sizes (Werner and Gilliam 1984, McCoy et al. 2011, Davidson et al. 2021). Growth as a defense against stage-specific consumers has been shown to be important mechanism for populations of amphibians (McCoy et al. 2011), fishes (Rice et al. 1993), terrestrial plants (Brice et al. 2024), as well as terrestrial and aquatic invertebrates (Pepi et al. 2018, Davidson et al. 2021). Research on size-structured interactions has historically focused on theoretical and empirical treatments of density-dependent growth rates, competition, ontogenetic habitat switching, population size-structure, and juvenile bottlenecks (Werner and Gilliam 1984, De

Roos et al. 2003). More recent studies of the interaction have focused on how individual growth rates alone influence cohort survival in size-structured populations (Craig et al. 2006, McCoy et al. 2011, Schmera et al. 2015, Brannelly et al. 2019), while only two studies have examined the theoretical effects of the interaction between juvenile growth and survival on populations dynamics (e.g., fish recruitment-Rice et al. 1993, equilibrium densities of caterpillars-Pepi et al. 2023). Explicit theoretical predictions about population growth rates are rare and no effort has been made to use theoretical predictions to interpret the combinations of field-measured demographic rates on population growth.

Size-indexed demographic models track size at age, combining juvenile growth rates and per-capita survival rates to make population growth projections. Such models can also be used to identify the demographic parameter space making population growth negative, zero, or positive and we suggest the parameter combinations can be profitably illustrated with zero population growth isoclines. Zero population growth isoclines have typically been used theoretically to predict population dynamics and/or coexistence outcomes for interacting species under variable parameter values and assumptions about the interactions (MacArthur and Levins 1964, Vance 1985), but zero-growth isoclines could also be derived numerically from demographic population models to identify parameter combinations producing zero population growth. To our knowledge this has not been done, but demographic isoclines from demographic models could offer quantitative maps for interpreting the interactive effects of juvenile survival and growth of a sensitive early life stage on population growth. Field measured parameters could then be compared to the isocline to identify natural spatial or temporal variation influencing recruitment or population growth. Here we describe examples from marine, terrestrial, and freshwater ecosystems that could be conceptualized using demographic isoclines and demonstrate the use of an isocline with a case study of a freshwater gastropod of conservation concern.

A demographic isocline can be created by identifying specific combinations of demographic rate parameters that produce populations at dynamic equilibrium ($\lambda = 1$) and plotting the rate combinations in demographic rate space. The two demographic parameters we use in this paper are juvenile survival which we define as the probability an individual of juvenile size will survive from one day to the next, and juvenile growth which we define as the daily change in size or mass. In contrast to juvenile survival which is a daily rate, cohort survival refers to the number of juveniles born in one year that survive to adulthood the next year which is the product of the juvenile survival and juvenile growth rate because juvenile growth rate determines the number of days a cohort experiences a survival rate in their sensitive stage. For a given life history, the hypothetical isocline in Figure 1A represents where births equal the juvenile losses ($\lambda = 1$) attributed to the product of juvenile growth and survival rates (i.e., cohort survival). Populations grow ($\lambda > 1$) when conditions produce demographic rates above and to the right of the isocline and decline ($\lambda < 1$) when average rate combinations fall below and to the left. The exact shape (i.e., steepness, linearity) of the isocline will depend on life history characteristics like longevity, adult survival, and fertility/reproductive rates that could be mediated by the environment or density-dependence. The stability of the isocline would depend on the relative constancy of the other life history characteristics of the population. If juveniles can escape vulnerable sizes through faster growth rates, then a negative slope would be expected (Figure 1A). In natural settings, spatiotemporal environmental factors that influence juvenile survival and growth will combine to mediate where the population exists in demographic rate space (Craig et al. 2006, Davidson and Dorn 2018, Davidson et al. 2021, Ma et al. 2021, Nunes et al. 2021, Meehan et al. 2022). We emphasize that the isocline is theoretical and identifies parameter values producing λ=1 from a size-indexed demographic model with other life stage parameters held constant. Density dependence in a real population could act at the juvenile stage and produce rates on the isocline, but should not fundamentally change the isocline. In contrast, if density-dependence acts on the adult stages it would conceivably affect the isocline (i.e, altering reproductive rates) and we discuss this in our case study, but do not explore the details further here. If most of the variation in juvenile survival can be attributed to predators (Werner and Gilliam 1984) then the locations of the rates in demographic state space indicate the degree of consumer control on the population and environmental mediation of such top-down control (Figure 1B-D).

The isocline approach could be of potential general interest, particularly in its application to populations with high recruitment variation. The potential utility of demographic isoclines is demonstrated here by depicting hypothetical isoclines for three taxa of conservation or management concern that span terrestrial, marine, and freshwater ecosystems where the details of recruitment have been sufficiently described (Figure 1B-D). In all three cases, consumption is the primary mechanism driving survival of individuals in juvenile stages, and the demographic parameters related to environmental gradients have been well studied. Thus, the qualitative predictions about where conditions fall relative to the isocline can be reasonably inferred (Figures 1B-D). By describing these three examples from the published literature under the same framework we do not intend to over-simplify the ecological details, but rather to conceptually illustrate the similar issue of the demographic rate combinations that can help population biologists, whether involved in conservation, resource management or pest management, assess the potential for population growth using a model and field-measured rates.

First, in a freshwater rock pool system a spatial temperature gradient simultaneously affects growth rates of larval mosquitoes (*Aedes atropalpus*) and their survival with dragonfly (*Pantala* spp.) naiad predators (i.e., per-capita foraging rate of the predator increases with temperature; Figure 1B; Davidson et al. 2021, 2024). The net effect of both rate changes was such that at cooler temperatures the mosquito populations would not recruit well even though survival was high because daily growth was too slow; rate combinations would be below and left of the isocline (Figure 1B). With higher temperatures the average daily survival decreased, but the increased daily growth rate with warmer temperatures shortened the time spent in the larval stage even further so that mosquito populations could recruit and grow; joint rates moved down, but also to the right of the isocline (Figure 1B). In a contrasting invertebrate system (not shown) of ant predator-caterpillar prey interactions in a terrestrial system, warmer temperatures affected both rates in the same manner, but increases in caterpillar growth with higher temperatures were unable to compensate for lower survival from ant predation (Pepi et al. 2018).

In a well-studied terrestrial ecosystem, recruitment (sucker to sapling transition) of quaking aspen (*Populus tremuloides*) that are browsed by elk (*Cervus canadensis*) are influenced by elk numbers and variation in moisture (Figure 1C; Brice et al. 2024). After wolves (*Canus lupis*) were extirpated from the Greater Yellowstone Ecosystem, USA (GYE) and elk abundance was high, the browsing pressure on aspen stands was high everywhere regardless of the available moisture, resulting in widespread recruitment failure (bottom of Figure 1C). Following wolf reintroduction the elk declined and browsing pressure was reduced (Kauffman et al. 2010, Brice et al. 2024), but aspen stand regrowth was variable (Ripple and Beschta 2007, Kauffman et al. 2010, Beschta and Ripple 2016). Recent evidence indicates that the patchy recruitment of aspen suckers from different stands may be from spatiotemporal variability in moisture allowing aspen suckers to grow to sizes large enough to escape the browsing pressure (upper points in Figure 1C; Brice et al. 2024). While no comparison of browsing rates and sucker growth has been conducted to our knowledge, this understanding of spatially patchy aspen recruitment in GYE was proposed by Kauffman et al. (2010) and the work by Brice et al. (2024) confirmed the importance of the interaction from the aspect of spatiotemporal variation in moisture. Examinations of the interactive effects of the two rates across spatiotemporal gradients, rather than being considered as alternative explanations (i.e., top-down vs. bottom-up factors), could improve the understanding of the patchy regrowth of aspen and other plants in response to predator reintroductions and herbivore densities (Beschta and Ripple 2016).

In a marine ecosystem, multiple environmental gradients (e.g., depth, salinity; Munroe et al. 2017, Baillie and Grabowski 2019) appear to affect one or both demographic rates of small settling oysters (e.g., *Crassostrea virginica* or *Saccostrea glomerata*) which are considered foundation species. Higher salinity produces faster growth of small oysters (Munroe et al. 2017), but high salinity can also facilitate outbreaks of predaceous drilling snails (Kimbro et al. 2013). How the two rates change together in state space with increased salinity is unclear from the literature, but salinity-mediated changes to survival and a subsequent population collapse (illustrated in Figure 1D) was observed by Kimbro et al. (2013) in an estuary.

While the specific slopes of the isoclines for these examples from freshwater, marine and terrestrial ecosystems are unknown, the qualitative description of the isocline should be generalizable (i.e., negative slope) to any species with high mortality from stage- or size-dependent juvenile predators. As a case study, we examined an annual gastropod of conservation concern, the Florida Apple Snail (*Pomacea paludosa;* hereafter FAS), to demonstrate the utility of a demographic isocline. We identify theoretical combinations of juvenile-stage parameters predicting population stasis (λ = 1), growth (λ > 1), or decline (λ < 1) from a previously parameterized size-indexed demographic model. Then, we quantified daily size- and season-dependent survival and juvenile growth rates in the field to interpret the combined effects that juvenile growth and survival have on predicted population growth. Using the isocline approach, the field measured values were interpretable from a population-growth perspective and provided material for generating novel hypotheses about population limitation.

Materials and methods

System and study species

The Florida Everglades is a shallow, expansive (~915,000 ha), subtropical, oligotrophic wetland covering much of southern Florida (Richardson 2010; Appendix S2: Figure S1). Rainfall is seasonal with approximately 80% of rain falling in the wet season from June–November (Gaiser et al. 2012). which produces intra-annual water depth fluctuations of ≥ 60 cm. The degree of water level recession and depth in the dry season is a function of rainfall and water management decisions. Historically, water flowed in a single shallow sheet from Lake Okeechobee at slow velocity across the spatial extent of the Everglades (i.e., sheet flow; Sklar et al. 2005), but flow was reduced or eliminated after compartmentalization and drainage. Compartmentalization and drainage of the Everglades altered the hydrologic conditions by increasing water depths in some areas but decreasing depths in others. Within the Everglades, the ridge-slough landscape originally covered 55% of the Everglades (McVoy et al. 2011), but now covers ~44% (Richardson 2010). In the ridge-slough landscape, ridges and sloughs differ slightly by elevation (~10–15 cm) which produces habitat/vegetation patterning. The lowest elevation slough habitats dry to sediment surfaces every 3–10 years and are dominated by floating vegetation like water lilies (*Nymphaea odorata*) or emergent spike-rushes (*Eleocharis* spp.).

Sloughs are interspersed with higher elevation ridges dominated by sawgrass (*Cladium jamaicense*) that dry most years (Zweig and Kitchens 2008). Ongoing hydro-restoration of the Everglades ecosystem aims to restore hydro-patterns to improve conditions for wildlife and natural communities.

The FAS is the largest native freshwater gastropod in North America, it inhabits shallow lakes and wetlands, and currently occurs at low adult densities (< $1/m^2$) in southern Florida wetlands (Gutierre et al. 2019). Snails grow from 3–4 mm shell length (SL) at hatching to > 40 mm SL as adults and do not live beyond 1.5 years (Hanning 1979). Most reproduction (~70%) occurs during cooler spring seasons when water levels are declining, and some reproduction occurs (~30%) during warmer early summer when water levels are rising (Hanning 1979, Barrus et al. 2023). At adult sizes (> 25 mm SL) FAS are a critical resource for the endangered Snail Kite (*Rostrhamus sociabilis*; Cattau et al. 2014), which experienced significant declines within the ridge-slough landscape from 2001–2010. The decline in Snail Kite populations in the Everglades is at least partly explained by declines in FAS, so improving the conditions within the ridge-slough landscape for FAS populations is imperative. As small juveniles (< 10 mm SL) FAS are prey for crayfish (*Procambarus* spp), sunfish, non-native cichlids, large killifishes (*Fundulus seminolis*), greater siren (*Siren lacertina*), and turtles (e.g., *Kinosternon bauri*; Valentine-Darby et al. 2015, Davidson and Dorn 2017). Giant water bugs (Belostomatidae, Kesler and Munns 1989) live in the Everglades, are known to eat snails but have not been studied. Juvenile FAS outgrow most common fish and invertebrate predators when they reach ~10–11 mm SL (Valentine-Darby et al. 2015, Davidson and Dorn 2017, Appendix 2: Figure S2). There has been no investigation of the factors that influence survival of juvenile snails after hatching or the population-dynamic impacts of cohort survival within natural wetlands mostly because tracking cohorts of small juvenile snails is logistically infeasible, thus we developed an isocline approach to investigate spatiotemporal variation in juvenile growth and survival using a demographic model.

Zero-Population Growth Isocline

We used a published size-indexed demographic population model (Darby et al. 2015) to create zero-population growth isoclines from theoretical combinations of two parameters, juvenile growth and survival (FAS < 10 mm SL) holding all other variables constant (details in Appendix S1). The model is an age-structured matrix model (i.e., Leslie Matrix) that tracks annual cohorts on daily time steps, but the model structure also tracks size on a daily step with an equation that describes size at age. Survival and fecundity rates are determined by the size-structure. Cohorts of juveniles are produced during the reproductive season (i.e., spring or summer) as a function of fecundity rates and water depths (i.e., depth dependent egg laying rates). The model was originally developed to provide ecological assessments of hydrologic restoration but we re-coded the model using R for our research application. We used most of the original parameters from Darby et al. (2015), but excluded the carrying capacity and included a few adjusted parameters to reflect recent changes in the understanding of FAS life history (see Appendix S1: Table S1). Numerical simulations were used to construct zero population growth isoclines by adjusting juvenile survival and growth parameters under two different hydrologic conditions which produced depth-dependent differences in reproduction (i.e., "Good Reproduction" or "Poor Reproduction"; Appendix S1). For each simulation, we used one juvenile growth rate, one juvenile survival rate, and one hydrologic condition, then we allowed the population to reach a stable size distribution. After a stable size distribution was achieved, we calculated the population growth rate (λ). After repeating this for 1666 different juvenile growth, survival and hydrologic conditions, we identified the parameter combinations that produced static population sizes λ = 1. We plotted these values as an isocline for each hydrologic scenario (For further details see Appendix S1). The isoclines graphically represent theoretical combinations of the two parameters that stop population growth (λ = 1). The isoclines are boundary conditions between a growing or a declining population assuming 1) a static hydrologic scenario and adult life history, and 2) that the juvenile growth and survival rates represent the average rate experienced by juvenile snails for the year. Because juvenile FAS densities are so low in our study wetlands (typically << 0.1/m^2) and yet juveniles can survive and grow to high sub-adult densities (16/m^2) in mesh predator-exclusion cages (Barrus et al. 2023), we considered negative density-dependent growth to be irrelevant to the rates the populations of FAS actually experience.

After constructing isoclines, we used *in situ* experimental techniques to calculate juvenile survival and growth parameters and their 95% confidence intervals and plotted them in demographic state space. The nature of the model made it impossible to change juvenile growth rates seasonally so the interpretations of the rates in state space assumed that the parameters were averages experienced throughout the year. The season-dependent predictions from the field measurements were, therefore, the expected snail recruitment given the rates measured for the respective season. Because ~70% of reproduction (egg laying) occurs in the dry season and ~30% occurs in the wet season (Darby et al. 2015, Barrus et al. 2023), we calculated the weighted averages of the seasonal parameters to obtain annual average values.

Survival and Growth in the field

We measured juvenile survival and growth in wetlands at the Loxahatchee Impoundment Landscape Assessment (LILA) and in two sites in the western portion of Water Conservation

Area 3A (WCA3A; Appendix S2: Figure S1) in Florida, USA. LILA consists of four 8 ha impounded wetlands with ridge and slough elevation features and hydro-patterns that mimic the wetlands of the Everglades (Appendix S2: Figure S1B). Water levels vary seasonally in the wetlands but the water levels in LILA were controlled by pumps and culverts to perform a landscape-scale hydrologic experiment. We worked in two wetland impoundments at LILA that had hydrologic conditions deemed good for FAS reproduction (Barrus et al. 2023). The two sites near the western boundary of WCA3A near Big Cypress National Park (Appendix S2: Figure S1: Sites 2 and 3 in Ruetz et al. 2005) were within a 1240 km^2 contiguous portion of the Everglades. The WCA3A sites were chosen because they were near locations of higher FAS densities in the recent past, including sites that previously supported Snail Kite nesting (Cattau et al. 2014).

We tethered snails by attaching monofilament to the apex of the shell using super glue, then attaching the other end of the monofilament to PVC poles within the wetland (Appendix S2). Tethered snails were placed on transects in the wetlands ~2 m apart and checked daily. Surviving snails were moved to increase independence between nights while depredated snails were replaced. Although in LILA we tethered snails of all sizes to test for size-dependence, here we focus on the survival of snails < 10 mm SL because this related to the isocline and was what varied the most seasonally (Appendix S2). We only tethered snails < 10 mm SL in the WCA3A. Further details of the tethering experiment can be found in Appendix S2. Here we also focus on relating survival to the isocline, but we also observed tethering artefacts of different predators that allowed us to identify common predators (see further discussion in Appendix S2).

Tethering can inflate mortality rates of animals capable of escape (Baker and Waltham 2020), but FAS are relatively less mobile than their typical predators and rely on retracting into their shell to avoid predation rather than escape. For prey with little escape capability, tethering should give reliable information about prey survival particularly across gradients of predation as it measures encounter rates (Rochette and Dill 2000, Ruehl and Trexler 2013). Prior experimental work with this species did not find any measurable anti-predator response, either morphological or behavioral, to chronic exposure to crayfish (Davidson and Dorn 2017).

We measured juvenile growth either using *in-situ* 1-m^2 mesh cages or with a regression that predicted wet season juvenile snail growth using total phosphorus (TP) concentrations in metaphytic mats ($R^2 = 0.85$; Barrus et al., 2023). The metaphyton (sometimes called periphyton) in the Everglades is a floating calcareous mat composed of algae, cyanobacteria, other microbes, and algal detritus (Gaiser et al. 2011). Algae was allowed to accumulate in the cages two weeks prior to the experiment, and two liters of metaphyton was placed inside the cages as a food source (Drumheller et al. 2022, Barrus et al. 2023). Juvenile snails were individually marked and placed in cages to grow for 4–5 weeks. We placed 8 cages in LILA during both seasons and 3 cages in WCA3A site 2 in the dry season. To estimate wet season juvenile growth at WCA3A site 3, we measured the TP of metaphytic mats to predict FAS growth using regressions from (Barrus et al. 2023). We were only able to obtain dry season growth rates for site 2 in WCA3A because low dry season water depths at site 3 made use of cage mesocosms impossible. Using the growth results we then calculated the growth parameter k_{growth} to relate the results to the isocline. K_{growth} is a measure of size-dependent daily growth rates that can be calculated from knowing the initial size, the final size and the maximum size. The maximum size was assumed to be 50 mm SL. Details on calculating the growth parameter (k_{growth}) can be found in Appendix S2.

Results

Zero-population growth isoclines created from the size-indexed age-structured population model produced a declining isocline consistent with the expected interaction between daily juvenile growth and survival rates (Figure 2). Combinations of the two parameters above and to the right of the isocline predict growing populations ($\lambda > 1$) while combinations below the isocline predict declining populations ($\lambda < 1$). The negative slope of the isocline illustrates that populations with faster juvenile growth can withstand lower daily survival. Populations experiencing slower juvenile growth rates will need higher juvenile survival rates to persist or grow ($\lambda \geq 1$). Hydrologic conditions that improved reproductive rates (i.e., eggs laid/female) moved the isocline down and to the left (gray isocline in Figure 2), making the population slightly more resilient to lower juvenile survival (e.g., withstanding 3.1% lower survival at juvenile growth of $k_{growth} = 0.07$) and/or slower juvenile growth (e.g., withstanding7.7% slower juvenile growth at juvenile survival rates of 0.80; Figure 2). The separation between the isoclines was greatest for combinations of faster juvenile growth and lower juvenile survival (Figure 2).

Empirical Juvenile Survival and Growth related to the Isocline

We observed variation in the measured juvenile survival and growth parameters across sites and seasons (Figure 3). Tethering snails from hatchling to adult sizes indicated that survival was strongly size-dependent in the dry season with snails < 10 mm SL heavily depredated (Appendix S2: Figure S2, Table S1). The shell artefacts of deceased snails (< 10 mm SL) suggested that the predators were primarily native invertebrates, *Belostoma lutarium* and *Procambarus fallax*, plus salamanders (Appendix S2). Predator surveys indicated that abundances were also variable between seasons and sites (Appendix S2).

Across both field sites the juvenile growth was faster in the warmer wet season than the dry season (Figure 2, Appendix S3: Figure S2). The dry season had lower juvenile survival and slower juvenile growth with combinations falling below and to the left of the isocline (Figure 2). In contrast, the wet season had higher juvenile survival and faster growth; with average combinations falling on the isocline (LILA wetlands) or even above and to the right (WCA3A site 2; Figure 2). Snails in WCA3A site 2 had faster juvenile growth than those in LILA (Figure 2). The combined effects, weighted by seasonal egg laying distributions, resulted in average juvenile survival and growth parameters that predicted a declining population for both wetlands, though confidence intervals slightly overlapped the zero-population growth isocline for WCA3A site 2 (Figure 2). The overlap of the confidence interval with the isocline, indicating potential population stasis, could only be observed for an isocline reflecting good hydrologic conditions for egg-laying in WCA3A site 2 (Figure 2).

Discussion

Using a size-indexed age-structured population model we produced zero-population growth isoclines illustrating the interactive effects of juvenile growth and survival of a predator-sensitive juvenile stage on population growth. The expected effect that faster juvenile growth can offset higher mortality was illustrated. Our work was specific to an annual freshwater gastropod with size-dependent survival, but the approach is conceivably applicable to any size-structured consumer-resource interaction. For example, field combinations of both demographic rates for a marine shrimp were examined for multiple populations by Chockley et al. (2008), but they were not compared against population dynamic predictions. The isocline from the demographic model allowed us to interpret field measured rates and conclude that the populations should be static or declining. Seasonal parameters further indicated that both juvenile survival and growth were poorer in the dry season (spring: Feb–April) which overlaps with most of the reproductive period of FAS. The results produce novel hypotheses about environmental variation and predator control that might limit the FAS in the Everglades. Creating similar demographic isoclines for other species could offer insights into the spatiotemporal conditions producing population growth and recruitment.

Growth-Survival Isocline

Our zero-population growth isocline was created using a specific model for an annual gastropod of conservation concern which presents some limitations to its generality. Future theoretical work with a general size-indexed demographic models should be explored to determine how other parameters might influence the shape of the demographic isocline. While density dependence can influence population dynamics acting on juvenile demographic rates (Accolla et al. 2024), it is important to remember that our demographic isocline is theoretical and represents rates where births equal the juvenile losses ($\lambda = 1$) attributed to the product of juvenile growth and survival rates (i.e., cohort survival) assuming all other life history parameters in a matrix are held constant. Thus, density-dependence acting on juvenile growth and survival of real populations should not affect the shape of a theoretical isocline, but only the actual empirical juvenile rates observed. In contrast, density-dependence or environmental factors affecting birth rates (e.g., adult fecundity or longevity) could alter the shape of the isocline. Our exploration for the FAS demonstrated that water depth variation that influenced egg laying altered the slope of the isocline (Figure 2). Exploring the factors that change the shape of demographic isoclines would be a fruitful area for more general exploration. While we built our theoretical isocline assuming density-independent population growth we emphasize that this does not assume the actual juvenile rates are density-independent. Our isocline does not claim anything about the relative importance of density dependence on population growth. Rather, density-dependence acting on one or both juvenile rates could be an explanation for where the rates exist in state-space. It is the job of the investigator to determine what drives variation in the two rates. If juvenile density dependence is expected to be an important driver, then researchers will need to measure growth and survival rates at relevant densities. Because current densities of FAS are extremely low compared to what the environment can support when predators are excluded (Barrus et al. 2023), we did not consider negative density dependence at the juvenile stage to be a good explanation for the demographic rates we observed in the field.

The negative slope of the FAS isocline was a qualitative prediction from our model evaluation and is the expected result from extensive empirical studies that have explored how growth mediates mortality rates from consumers as juveniles pass through a sensitive stage (Jeyasingh and Weider 2005, Craig et al. 2006, Davidson and Dorn 2018, Pepi et al. 2018, Ma et al. 2021, Meehan et al. 2022, Davidson et al., 2024). The negative slope of the isocline result was consistent for different hydrologic conditions that mediate reproductive output of the adult snails (Figure 2), including hydrologic conditions that were held constant at the ideal depth for reproduction (Appendix S1: Figure S3). Additionally, we included three examples which growth and survival rates were sufficiently studied in the field to highlight their potential for applying this approach (Figure 1). Studies of the theoretical interactions between juvenile growth and survival scaling to population growth remain rare; our literature search found only one theoretical study exploring fish recruitment (Rice et al. 1993), and one theoretical study exploring how equilibrium densities of predator and prey populations are affected by juvenile growth and survival (Pepi et al. 2023). Although shown in a specific case, we expect the negative slope of the isocline would hold for any species with populations experiencing high recruitment variation that is driven by consumers.

When measuring survival and growth in the field, the primary interest is where the parameters fall relative to the isocline. And making specific predictions about a species of conservation concern *requires* a specific model for that species. Demographic models are becoming a widespread tool for conservation and ecology (Gacoigne et al. 2023), to the best of our knowledge, our work is the first to demonstrate their use in interpreting survival and growth rates measured in natural environments. Employing demographic isoclines in population research would require the development of a size-indexed demographic model that allows the isolation of the growth and survival of sensitive juvenile stages. The time steps in the model should be short relative to the length of the sensitive stage and the measurements in the field should be measured commensurately to match the theoretical predictions.

Interpreting empirical measures of survival and growth

Including an isocline analysis of survival and growth allowed us to interpret natural empirically-measured parameters in a population dynamic perspective and offers insights about how environmental variation might influence consumer-resource interaction strength (Davidson et al. 2021, Pepi et al. 2023). Recent studies of temperature-dependent rates concluded that consumer-resource interaction strength should weaken or strengthen depending on asymmetries between thermal responses of the resource growth rate and consumer per-capita foraging (Davidson et al. 2021, Pepi et al. 2023). For our wetlands, the per capita foraging rates of ectothermic predators increased in the warmer wet season (calculation in Appendix S2: Figure S4), which should strengthen interactions between FAS and their predators (Figure 2) except that predator abundances after the wetlands reflooded decreased between seasons (Appendix S2: Figure S4). The reduction in predator abundances appears to have overwhelmed any increase in in per capita consumption that was mediated by warmer temperatures (Appendix S2). Studies that isolate the effects of variable environmental conditions on predator-prey interactions have typically controlled predator densities experimentally, or statistically (Jeyasingh and Weider

2005, Davidson and Dorn 2018, Pepi et al. 2018, Davidson et al. 2021, Ma et al. 2021) to measure per-capita rates. But the study of natural populations in communities requires consideration of natural seasonal variation in predator abundances that could covary with temperature (Davidson et al. 2024). Environmentally-mediated changes in predator communities may be more important to survival than the per-capita foraging rates of the predators and could either counteract or exacerbate the temperature-mediated changes to per-capita foraging rates.

Novel Hypotheses for Population Management

The hydrologic conditions within the Everglades are heavily managed with the goal of restoring conditions for wildlife and fishes, historic habitat mosaics, and biodiversity, while safe-guarding drinking water (National Academies of Sciences, Engineering, and Medicine 2021). Improving conditions for the FAS populations in the Everglades will be necessary to delist the federally endangered Everglades Snail Kite. The current paradigm for encouraging population growth of the FAS is to make hydrologic conditions more favorable for reproduction (i.e., maintaining water levels in a shallow range of depths that encourages egg laying; Darby et al. 2015), but our results indicate that with the current levels of predation and individual growth, improving hydrologic conditions for reproduction in the Everglades can, at best, only maintain the already small populations of FAS. For FAS population growth to become positive, we offer three hypotheses (Table 1) about the spatiotemporal environmental conditions that could shift the average daily survival and growth conditions experienced by juveniles.

The dry season parameters were worse than the wet season parameters which is particularly problematic because most egg-laying occurs during the dry season (spring) before the water reaches its annual minimum depth (typically in May; Barrus et al. 2023). This result suggests that FAS may benefit more by improving dry season conditions for survival and growth than by improving wet season conditions. If females can store their resources and hydrologic conditions could conceivably shift more of the egg laying to the wet season, then the average demographic parameters would move up and right towards stasis or growth (Hypothesis 1-Table 1). Although more research is needed to understand how water levels might mediate this response, one observation suggests that shifting reproduction to the wet season (July–August) can occur in shorter-hydroperiod wetlands outside the ridge-slough landscape (O'Hare 2010).

Improved food quality could also move parameters to the right in state space (Hypothesis 2-Table 1). The Everglades is phosphorus-limited with natural oligotrophic total phosphorus concentrations between 110–400 $\mu g \cdot g^{-1}$ in the ridge and slough landscape (Gaiser et al. 2011). Growth of juvenile FAS varies with TP concentrations in the periphyton (Hansen et al. 2022, Barrus et al. 2023), and previous experimental manipulations of phosphorus showed that higher TP increased growth and juvenile apple snail survival in the presence of crayfish (Davidson and Dorn 2018). Our results build on this finding by indicating that TP can mediate the net community level effects of predators on recruitment in the field. Periphyton TP levels were highest at WCA3A site 2 (Table S3), and it was the only site that had wet season growth and survival parameter combinations that predicted population growth. Nevertheless, restoration and management actions attempt to avoid eutrophication of the Everglades (National Academies of Sciences, Engineering, and Medicine 2021). Perhaps more promisingly, the pre-drainage Everglades was a flowing ecosystem (the "River of Grass") with water velocities > 2 cm/s. Flow velocity was shown to increase growth of *Pomacea* apple snails through changes to microbial food quality (Hansen et al. 2022), and therefore restoration of flowing water might improve growth rates of juvenile FAS (Hypothesis 2-Table 1).

Finally, the predation rates in the Everglades might currently be higher than historical levels as a function of non-native fishes (Pintar et al. 2023) or hydrologic conditions supporting higher densities of juvenile invertebrate predators in the sloughs (Hypothesis 3-Table 1). Some non-native fishes introduced to the Everglades have molluscivorous tendencies, but our observations of diets and tethering remains suggested that native predators (e.g., crayfish, giant water bugs, greater sirens) in LILA seem to be more responsible for mortality patterns than non-native species like Mayan cichlids (*Mayaheros uropthalmus*). Because the predator assemblage feeding on juvenile snails included native species existing across a wide range of the hydroperiod gradient, it remains unclear how water depths or hydrologic droughts would shift survival rates (e.g., drying benefits crayfish populations; Dorn and Cook 2015, Sinnickson and Dorn 2024).

Conclusion

We demonstrated the first demographic isocline approach that could be used when studying and managing size-structured consumer-resource interactions. The demographic isocline was conceptually derived from well-studied principles of size-structured interactions (Werner and Gilliam 1984, Davidson et al. 2024), and the operationalized isocline was built using a size-indexed demographic model and theoretical combinations of daily survival and growth. The isocline illustrated the expected negative relationship between juvenile growth and survival; populations with faster-growing juveniles can withstand (λ=1) greater mortality (lower survival). A demographic isocline can be used as a map to interpret empirically measured rates in a population dynamic context. Our case study indicated that seasonal changes in the predator community were more important in determining interaction strength than simpler physiological expectations based on thermal responses of predators and prey. This demographic isocline approach provided novel hypotheses about the conditions needed to restore a historical resource of an endangered species. We encourage others to consider further theoretical exploration of the concept and development of size-indexed models and demographic isoclines to study populations with high recruitment variability.

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Author Contributions

All authors contributed to the design. The tethering and growth rate design and experiments were established by NJD, NTB and MIC. Development of the isocline was conducted by NTB in consultation with NJD. Data collection was performed by NTB and NJD. Analyses and statistical models were conducted by NTB in consultation with NJD. The paper was written by NTB and NJD with edits from MIC. All authors have read and approved the final manuscript.

Conflict of Interests

We declare no financial interests that could create conflicts for this work.

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Tables

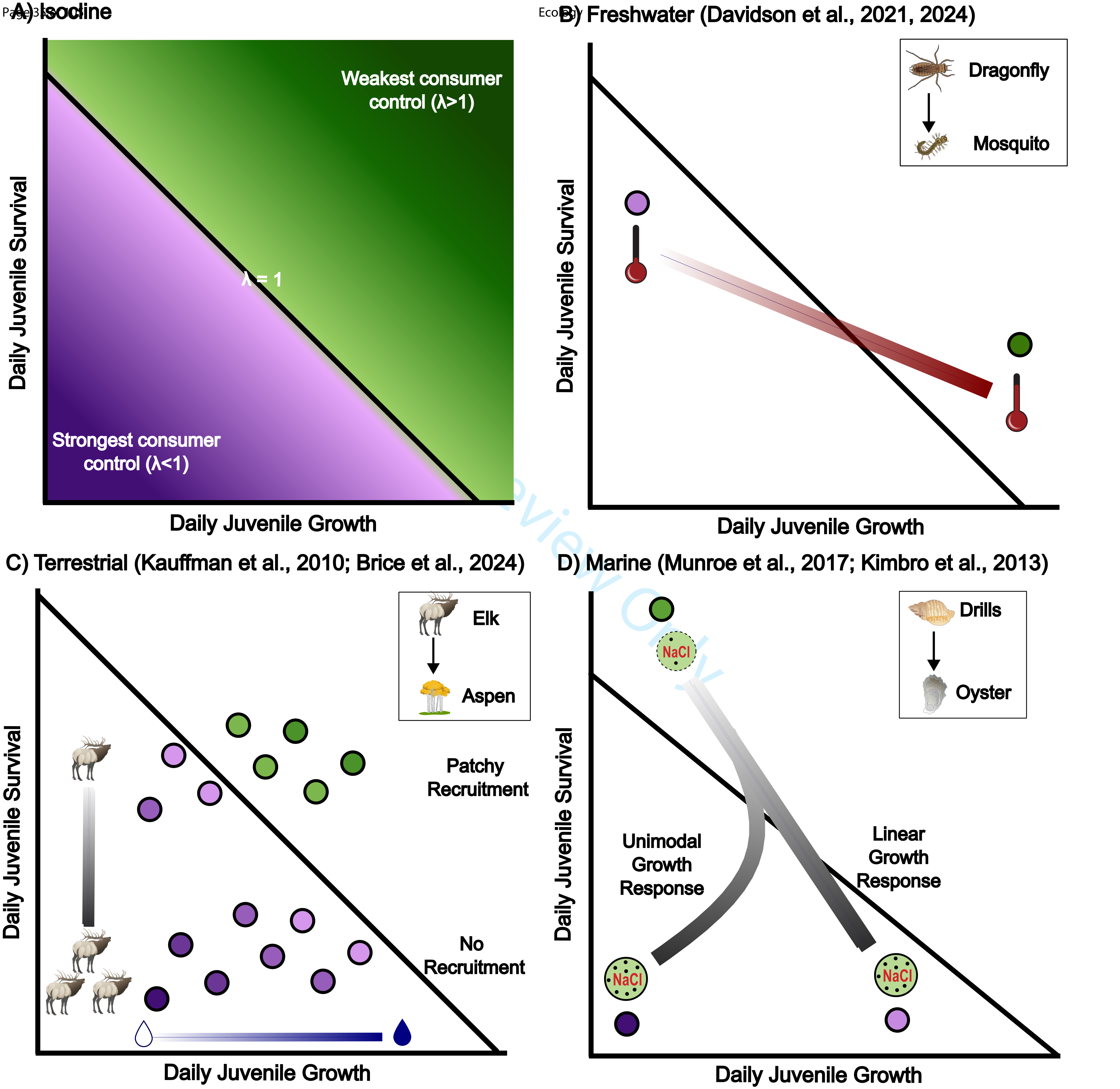
Table 1: Three hypothesized changes in spatiotemporal conditions that could shift juvenile FAS survival and growth parameters from the left (declining) side of the isocline to the right side of the isocline.

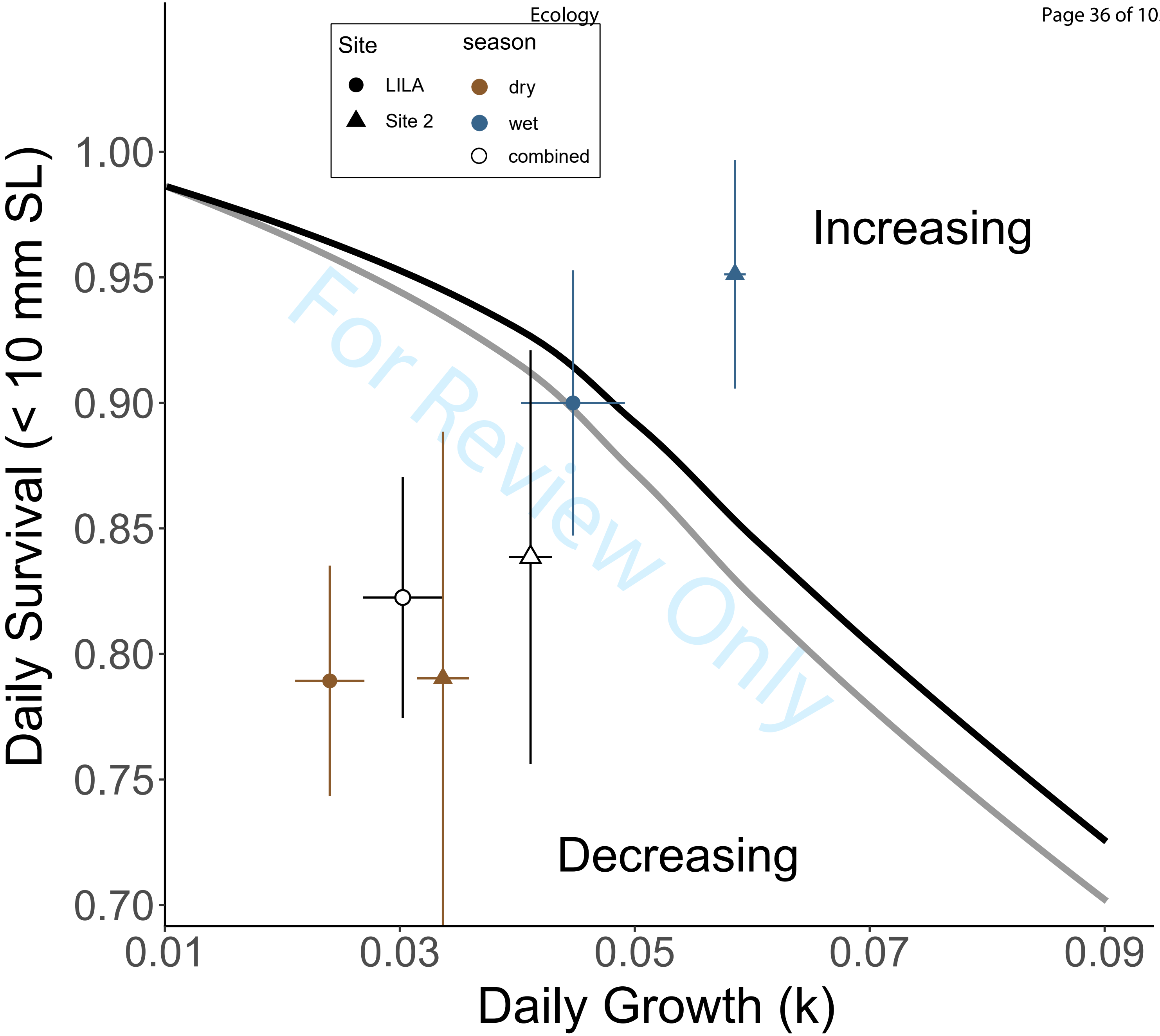
Hypothesis	Predicted Parameter Shift	Description of Environmental Change Hypothesis
1	Up and right	Seasonal depth/hydro-patterns that shift more of the egg production to the wet season expose more juveniles to favorable wet season parameters.
2	Right	Hydrologic conditions that improve growth rates (e.g., flow or flow-loading of nutrients) through improved food quality or water quality (oxygenation).
3	Up	Hydrologic patterns that disfavor important predators could improve survival for juvenile snails (especially in the dry season).

Figure Captions

Figure 1. A) Zero-population growth isocline illustrating the expected joint impact of growth rates and mortality to consumers at a sensitive stage (diagonal black line). Resource populations are decreasing/recruiting in areas to the left and below the isocline shaded purple) while areas above and to the right indicate populations that are increasing/recruiting (shaded green). The strength of consumer control is represented by the gradient from dark green to dark purple. B-D) Examples of populations spanning freshwater, terrestrial and marine ecosystems for which hypothetical demographic isoclines could provide conceptual meaning to population dynamics and predictive values for field measured rates. The color of points corresponds to their position along the gradient in A). B) The relationship between mosquito (*Aedes atropalpus*) population with predatory dragonflies (*Pantala* spp.) depends on joint temperature mediation of survival and growth of mosquitos. C) Quaking aspen (*Populus tremuloides*) stand recruitment depends on moisture/precipitation gradients and elk (*Cervus canadensis*) browsing rates driving sucker survival. D) Oyster (*Crassostrea virginica*) growth and mortality to predatory drilling snails are affected by salinity gradients, though growth rate declines as seen from low to intermediate level have not been described at high salinity levels and could follow linear or unimodal relationships. All vector graphics except the dragonfly and mosquito larvae are from the University of Maryland's Integration and Application Network (ian.umces.edu/media-library). The dragonfly and mosquito larvae are from Aquatic Ecology Lab at Florida International University (FIU), principal investigator Nathan J. Dorn.

Figure 2 Isoclines illustrating the interactive effects of juvenile growth and survival that produce zero net population growth for a size-structured model of a freshwater gastropod (*Pomacea* *paludosa*) under two different hydrologic regimes that affect reproduction (black isocline = lower reproduction, gray isocline = higher reproduction). Points are mean daily survival (snails < 10mm SL) and growth (k_{growth}) quantified in LILA wetlands and site 2 in Water Conservation Area 3A. Error bars represent 95% confidence intervals for each parameter estimate. The combined parameters (open symbols) were calculated by a weighted average reflecting greater juvenile snail production (egg laying and hatching) in the dry season.


A) Isocline
Weakest consumer control (λ>1)
λ = 1
Strongest consumer control (λ<1)
Daily Juvenile Survival
Daily Juvenile Growth
B) Freshwater (Davidson et al., 2021, 2024)
Dragonfly
Mosquito
Daily Juvenile Survival
Daily Juvenile Growth
C) Terrestrial (Kauffman et al., 2010; Brice et al., 2024)
Elk
Aspen
Patchy Recruitment
No Recruitment
Daily Juvenile Survival
Daily Juvenile Growth
D) Marine (Munroe et al., 2017; Kimbro et al., 2013)
Drills
Oyster
NaCl
Unimodal Growth Response
Linear Growth Response
NaCl
NaCl
Daily Juvenile Survival
Daily Juvenile Growth


Site
LILA
Site 2
season
dry
wet
combined
Increasing
Decreasing
Daily Survival (< 10 mm SL)
1.00
0.95
0.90
0.85
0.80
0.75
0.70
0.01
0.03
0.05
0.07
0.09
Daily Growth (k)
For Review Only

Journal: Ecology

Article Title: Interpreting field measurements of juvenile growth and survival rates with population growth isoclines

Authors: Nathan T. Barrus, Mark I. Cook & Nathan J. Dorn.

Appendix S1: Isocline Development

We used a published stage-structured model called EVERSNAIL (Darby et al., 2015) (hereafter referred to as 'the population model') to identify juvenile survival and developmental rate parameters that are expected to produce growing populations of apple snails. The population model was created to project population size across the extent of the Everglades and includes local scale sub-models that parameterize life history (i.e., survival, developmental rates, and reproduction). Depth and temperature data used in the population model from the Everglades was provided from the Everglades Depth Estimation Network (EDEN; Jones, 2015) and South Florida Water Management Districts online database (DBHydro; www.sfwmd.gov/science-data/dbhydro), respectively. The population model was built with the best available understanding of the Florida Apple Snail life history and includes life history responses to hydrologic variation. The model projects age- and size- structure on a daily time step using matrices (a.k.a. Leslie Matrix). The structure of the matrices are based on size which are indexed using a growth at age equation. Survival and fecundity rates within the transition matrix are size-dependent. The transition matrix and structure matrix are given below:

$$A = \begin{vmatrix} 0 & 0 & 0 & . & f_i & f_{i+1} & f_{i+2} & . & f_{499} & f_{500} \\ a_{12} & 0 & 0 & . & 0 & 0 & 0 & . & 0 & 0 \\ 0 & a_{23} & 0 & . & 0 & 0 & 0 & . & 0 & 0 \\ . & . & . & . & . & . & . & . & . & . \\ 0 & 0 & 0 & . & 0 & 0 & 0 & . & 0 & 0 \\ 0 & 0 & 0 & . & a_{i,i+1} & 0 & 0 & . & 0 & 0 \\ 0 & 0 & 0 & . & 0 & a_{i+1,i+2} & 0 & . & 0 & 0 \\ . & . & . & . & . & . & . & . & . & . \\ 0 & 0 & 0 & . & 0 & 0 & 0 & . & 0 & 0 \\ 0 & 0 & 0 & . & 0 & 0 & 0 & . & a_{499,500} & 0 \end{vmatrix} \text{ and } N = \begin{vmatrix} N_1 \\ N_2 \\ N_3 \\ . \\ N_i \\ N_{i+1} \\ N_{i+2} \\ . \\ N_{499} \\ N_{500} \end{vmatrix}$$

Where A is the transition matrix from one daily step to another, N is population vector based on age. $a_{i,j}$ are the age specific daily survival probabilities, and f_i are the age specific fecundity rates. These matrices provide the age index for the population size structure which is determined by the following size at age equation.

$$Size_i = \frac{Size_{min} e^{k_{growth} Age}}{1 + (Size_{min} / Size_{max})(e^{k_{growth} Age} - 1)}$$

$Size_{min}$ = linear size of newly hatched snail, $Size_{max}$ = maximum size of snail, and k_{growth} = daily growth rate. Survival during hydrological droughts and depth-dependent reproduction tie the model to hydrologic variation (Darby et al., 2015). For further details see the appendix in Darby et al., (2015).

We wanted to use the model to examine individual juvenile stage parameters and at a local scale, so we re-coded the population model for research in R version 4.0.3 using the parameter details found in the supplement (Darby et al., 2015). While most of the parameters were left as described by the original model (Table S1.1), two parameters were altered. First, the number of egg masses produced per female was changed by standardizing reproductive effort across the life span of a female snail. A maximum number of egg masses that a female can produce was discussed in a large unpublished review of apple snail ecology (Pomacea Project, 2013); to standardize reproductive output, the population model's current parameter (Mass Size) was multiplied by the maximum number of egg masses a female can lay and then divided by the life span of the female (500 days in the model). Second, we removed the carrying capacity from the model to examine what parameter values allow the population to increase.

Four parameters were used to model developmental rates and juvenile survival. Developmental rates were determined by the parameter k_{growth} and assumes size- dependence. The initial parameter estimate for k_{growth} in the population model was 0.05. There were three parameters ($Surv_1$, $Surv_2$ and $Surv_3$) that determined juvenile survival during wet condition and were based on size classes ($Surv_1$ = 3-6 mm, $Surv_2$= 6-10 mm, $Surv_3$ = 10-16 mm SL). A fourth rate was used for large juvenile and adult snails ($Surv_4$ > 16 mm SL). Under the parameters in the population model, survival through the juvenile stage (3-16 mm SL) was constantly high (98.7% $\cdot$ day^{-1}). Survival slightly increased after snails reached 16 mm SL (99.0% $\cdot$ day^{-1}) and remained constant until the snails reached 500 days when survival declined to 0 reflecting adult senescence (Hanning, 1979). Alternate survival parameters were included in the population model for conditions of hydrological drought (dry sediment surfaces in the dry season), but the drought parameters were not important for our simulations.

To determine growth and survival parameters that controlled population growth, we calculated population growth through combinatorial re-assessments with different values under two different hydrologic regimes. We chose the wet condition parameters for survival to make the simulations most representative of the sloughs in the ridge-slough landscape. Before we started simulations aimed at varying developmental rate and survival parameters under different hydrologic regimes, we obtained an initial population size with a stable size structure. We ran the model using the model's original developmental rate and survival parameters for ten years of repeated depth and temperature data (January 1^{st} to December 31^{st}, 2020). The hydrologic data was taken from DBHYDRO's depth transponder in LILA's wetland M2, and the air temperature data was taken from the transponder nearest to LILA in West Palm Beach, FL (transponder coordinates: 26.6548^{0}N, 80.0669^{0}W). We tested differences between three starting hatchling numbers (100, 1000, and 10000 hatchlings), but starting numbers did not influence population growth.

Following this 10-year simulation to establish a stable size structure, we introduced two different hydrologic regimes repeated for 5 years that varied in depth-dependent egg-laying conditions. First, we used the poor reproduction hydrologic conditions from LILA that was deeper in the wet season of 2020 (Figure 2A). Next, (2) we used the good reproduction conditions (Figure 2A). The model runs with poor and good reproduction hydrographs were both conducted using natural temperature regimes taken from West Palm Beach, FL (Appendix 1).

Under each hydrological regime, simulations were conducted under different combinations of the parameters k_{growth}, $Surv_1$, and $Surv_2$. k_{growth} values were allowed to vary from 0.01 to 0.09 using increments of 0.005 and the two small juvenile survival parameters for wet conditions were decreased by 5%, 10% 15%, 20%, 30% and 40% of the starting values (0.987 day^{-1}). Simulations were run under all combinations of the variations in the three parameters ($n_{simulations}$ = 833 per hydrologic regime). The population size on every simulated February 1^{st} was taken to calculate an annual population growth rate (e.g., $\lambda_i = N_i/N_{i+1}$; where i = year). February 1^{st} was used because it corresponded to the day when the population model initiates the reproductive season. The geometric mean of the annual population growth rates over 5 years was taken to obtain a λ_{avg}. The intrinsic rate of increase (r) was then calculated by taking the natural logarithm of λ_{avg}. When r =

0 a population is at replacement, when r < 0 a population is declining, and when r > 0 a population is increasing.

The results of the simulations were used to identify combinations of development rates and survival of juveniles that determined thresholds (r = 0) for population growth given the two hydrologic regimes. Although the simulations were conducted with individualized parameters for the two size classes, we reduced dimensionality to aid in interpretation by multiplying the two juvenile survival probabilities which we named cumulative juvenile survival (i.e., survival < 10 mm SL = CJS; Figure 1A). At each level of k_{growth}, the intrinsic rate of increase (r) was regressed (Ordinary Least Squared-OLS) as a function of CJS, then the regression equation was used to solve for the CJS for which r = 0. The combinations of individual growth (k_{growth}) and juvenile survival (CJS) were plotted as zero population-growth isoclines (Figure 2B).

Table S1: List of parameters from EVERSNAIL, their values, the vital rate function they influence, what the function's purpose is in the population model, the adjusted parameters, and short description of the altered parameters.

Parameter	Value	Vital Rate F(x)n	F(x)n purpose	Adjustment	How
$Size_{min}$	3 mm	Growth 1	Model individual growth	No	
$Size_{max}$	50 mm	Growth 1	Model individual growth	No	
k_{growth}	0.05	Growth 1	Model individual growth	Alter	Explore values (0.01-0.09) and measure in the field
$Surv_1$ (3-6mm)	0.987 day^{-1}	Survival 1	Model size-dependent survival under wet conditions	Alter	Explore decreases of 5%-40% and measure in the field
$Surv_2$ (6-10mm)	0.987 day^{-1}	Survival 1	Model size-dependent survival under wet conditions	Alter	Explore decreases of 5%-40% and measure in the field
$Surv_3$ (10-16mm)	0.987 day^{-1}	Survival 1	Model size-dependent survival under wet conditions	No	
$Surv_4$ (>16mm)	0.99 day^{-1}	Survival 1	Model size-dependent survival under wet conditions	No	
$Surv_{drought1}$	0.976 day^{-1}	Survival 2	Model size-dependent survival under dry conditions	No	
$Surv_{drought2}$	0.984 day^{-1}	Survival 2	Model size-dependent survival under dry conditions	No	
$Surv_{drought3}$	0.989 day^{-1}	Survival 2	Model size-dependent survival under dry conditions	No	
$Surv_{drought4}$	0.99 day^{-1}	Survival 2	Model size-dependent survival under dry conditions	No	
Age_{mort}	500 days	Survival 3	Induce rapid die off of adults after 1.5 years old	No	
k_{age}	0.1 day^{-1}	Survival 3	Induce rapid die off of adults after 1.5 years old	No	
Mortality Threshold	27.5 mm	Survival 3	Induce rapid die off of adults after 1.5 years old	No	
Egg Mass Size	30 eggs	Reproduction 1	Give a measure of fecundity	Change	Standardize to eggs produced per female
k_{repr}	1	Reproduction 2	Model the relationship between fecundity and water depth	No	
$Depth_{mid}$	50 cm	Reproduction 2	Model the relationship between fecundity and water depth	No	
W_k	40	Reproduction 2	Model the relationship between fecundity and water depth	No	
$Depth_{min}$	10 cm	Reproduction 2	Model the relationship between fecundity and water depth	No	
$Depth_{max}$	90 cm	Reproduction 2	Model the relationship between fecundity and water depth	No	
k_{temp}	1 degree C^{-1}	Reproduction 3	Model the relationship between fecundity and temperature	No	
Temperature Threshold	17 degree C	Reproduction 3	Model the relationship between fecundity and temperature	No	
Female	0.5	Reproduction 4	Females alone can lay eggs	No	
Peak Reproduction	1 (Feb-June)	Reproduction 5	Model seasonal effects on fecundity	No	
Minor Reproduction	0.3 (June-Sep)	Reproduction 5	Model seasonal effects on fecundity	No	
No Reproduction	0 (Sep-Feb)	Reproduction 5	Model seasonal effects on fecundity	No	
Carrying Capacity	35000 egg masss ha^{-1}	Reproduction 6	Provides density dependence so the population cannot grow towards infinity	Remove	Explore threshold of increasing and decreasing populations

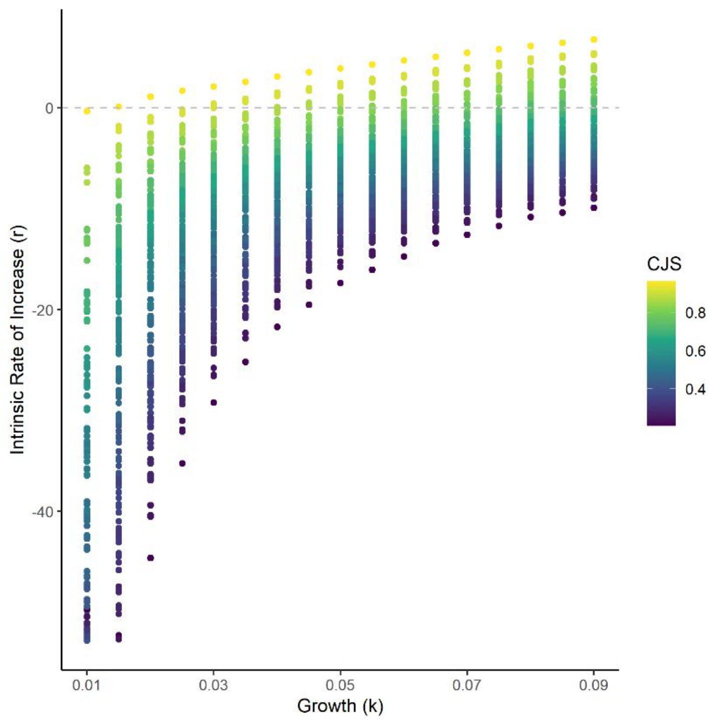


Figure S1**:** Scatterplot showing the intrinsic rate of increase (r) as a function of k_{growth} and Cumulative Juvenile Survival (CJS) from all simulations. The dashed line indicates an r = 0 which means populations are at replacement (i.e., not increasing nor declining).

Maximizing reproduction for isocline

To determine if maximizing reproduction could shift population growth to increasing, we created an isocline under constant depth and temperature conditions that maximizes reproductive effort.

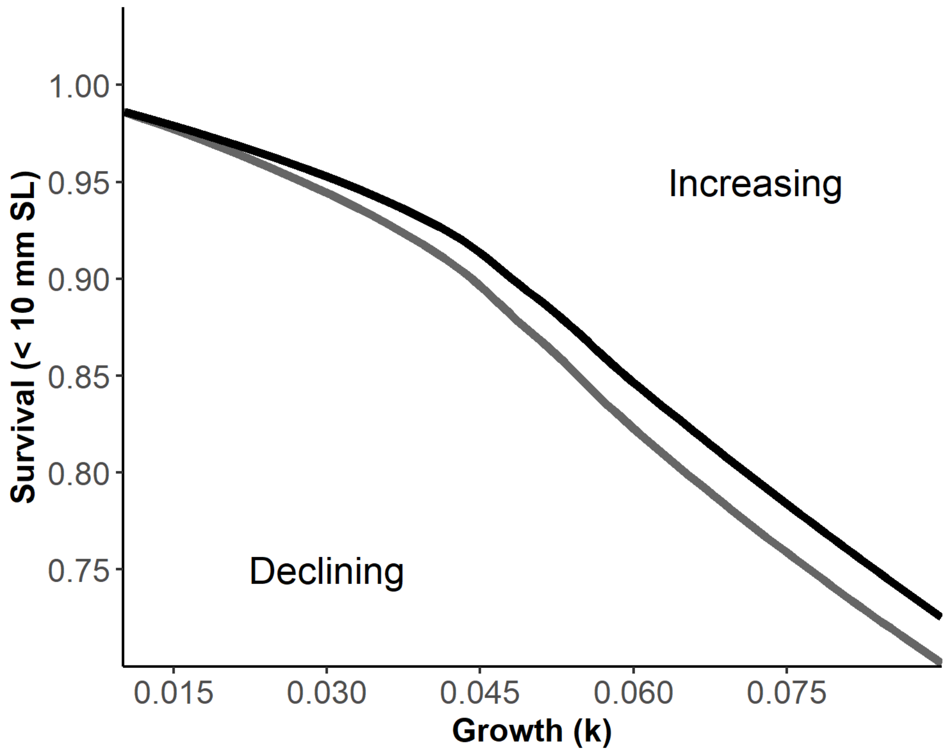


Figure S2: Isoclines illustrating the bivariate effects of juvenile growth and survival that produce zero net annual population growth for a size-structured model of a freshwater gastropod (*Pomacea paludosa*) under different hydrologic regimes that affect reproduction. The black isocline and gray isoclines represent three hydrologic scenarios producing maximized natural better (Dark Grey) and natural worse (Black) reproductive conditions. Better hydrologic conditions shift the isocline down and to the left enabling populations to withstand lower survival and slower growth.

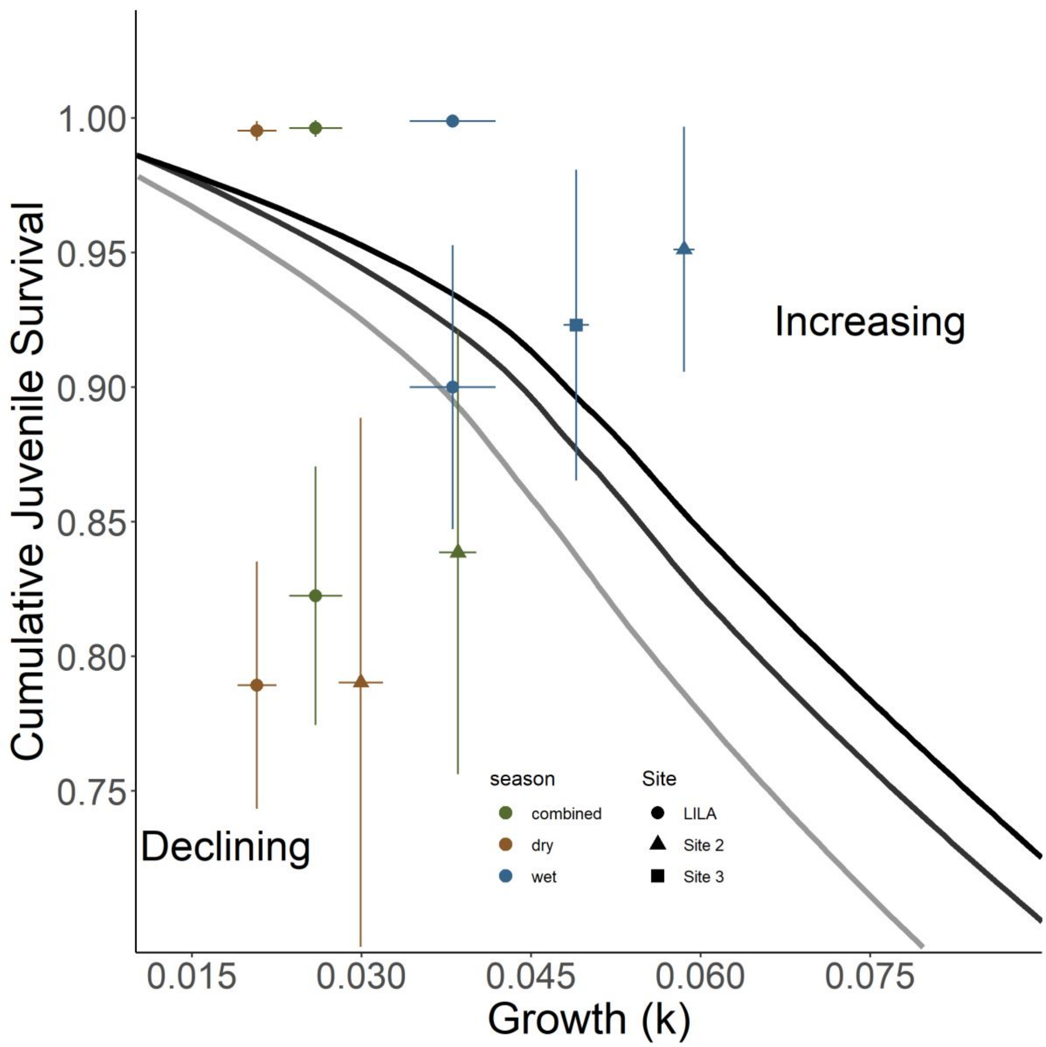


Figure S3: Isoclines illustrating the bivariate effects of juvenile growth and survival that produce zero net annual population growth for a size-structured model of a freshwater gastropod (*Pomacea paludosa*) under different hydrologic regimes that affect reproduction. The black isocline and gray isoclines represent three hydrologic scenarios producing maximized (Light Grey) natural better (Dark Grey) and natural worse (Black) reproductive conditions. Mean survival (snails < 10mm SL) and growth (k_{growth}) quantified in LILA and WCA3A are plotted on each panel with seasonal and combined parameters. The combined parameters were calculated by a weighted average reflecting greater juvenile snail production in the dry season.

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Journal: Ecology

Article Title: Interpreting field measurements of juvenile growth and survival rates with population growth isoclines

Authors: Nathan T. Barrus, Mark I. Cook & Nathan J. Dorn.

Appendix S2: Survival and Growth

Study map

We took empirical measuments of daily survival and growth in the wet and dry season within the experimental wetlands of the Loxahatchee Impoundment Landscape Assessment (LILA) and sites within everglades as shown in Figure S1.

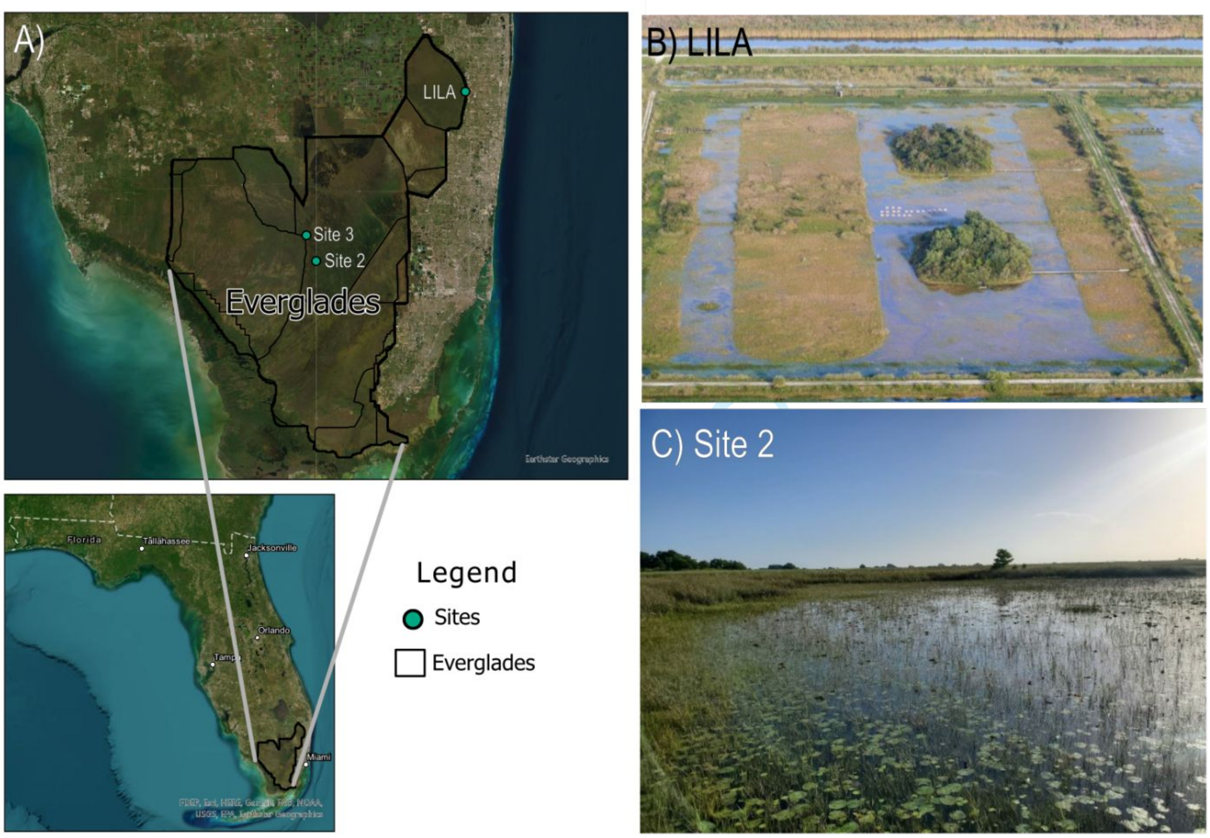

Figure S1 A) Map and images of B) LILA impoundment #2 and C) Site 2 in Water Conservation Area 3A. Photo credits to B) Mark I. Cook and C) Nathan T. Barrus.

Tethering

We conducted tethering experiments to measure survival of snails < 10 mm SL in LILA and in WCA3A each season to relate to the zero-population growth isocline. The experiments in the LILA wetlands also allowed us to test for size-dependent survival more broadly. We tested size- and season-dependent survival in two wetlands in LILA by tethering lab-reared juvenile snails from hatchling to adult sizes (3-30 mm SL) each season and measuring 24 h survival. In WCA3A, we only tethered juvenile snails (3-10 mm SL). Each tethering experiment was conducted by attaching snails to PVC poles with monofilament line on transects within the sloughs (Figure 3). The transects attempted to capture potential spatial variation in survival and were arranged "near" or "far" from the ridge edge (~5m and 15-20m, respectively). Across transects, tethered snails were placed apart to increase spatial representation and independence (Figure 3). We included 5-10 replicates of 3-mm size increments (i.e., 3-6mm, 6-9mm, 9-12mm,12-15mm, 15-18mm, 18-21mm, and >21mm SL) on each transect in LILA and 10-15 replicates of each 3-mm size increment (i.e., 3-6mm, 6-9 mm, >9 mm) in WCA3A. Snails were tethered by gluing 20 cm of either 2.4 lb (FAS ≤6mm SL) or 4 lb (FAS ≤6mm SL) monofilament line to the shell apex. Poles were placed ≥2 m apart and additional methodological details and the spatial considerations can be found in the supplement.

Tethering experiments were run for two-three days and snail status was checked daily. We checked snail status by prodding the operculum to incite movement, and we scored the status by five categories: (1) "missing" if the snail was removed from the tether, (2) "crushed/peeled" if the tether had shell fragments remaining on the tether, (3) "empty" if the soma from the shell had been removed, (4) "dead" if snails did not respond when prodded and (5) "alive" if snails responded when prodded. Using the snail status measures, snails that were "alive" were counted as survivals, while snails that were deemed "missing", "crushed", "dead", or "empty" were counted as mortalities. Surviving snails were placed back onto PVC poles and mortalities were replaced with tethered snails of the same size. To generalize measured survival to a larger area than the initial locations, tethers were moved two meters in a randomly chosen cardinal direction to increase independence between nights. The fate of each snail-day combination was considered an independent measure of daily survival. We ran the tethering experiments to achieve ~ 30 observations of mortality per size class. To ensure that snails could not escape tethers, tethered snails within each size class were caged in LILA for 72 hours to exclude predators. No snails escaped or died on tethers during 72 hours in the cages.

We analyzed the tethering dataset from LILA that tethered the full-size range of snails using logistic regression to test for size and season dependence of daily survival. We modeled survival using length (SL mm), transect ("near" or "far"), wetland ("M2" or "M4"), and season ("wet" or "dry") as covariates. We created a list of logistic models that included all possible combinations of these covariates and their two-way interactions (Table S2). Higher order interactions were excluded. The resulting models were compared using AIC scores, the structure of models with ΔAIC < 4 were examined, and the most supported model (lowest AIC) was selected for interpretation and evaluation. Logistic regression was fitted using the "glm" function in R v4.0.3 (R Core Team, 2023).

Overall, we observed a total of 759 independent observations of survival across two wetlands and two tethering seasons in LILA. After 24 hours, 654 snails survived, 43 snails were missing, 31 snails were empty, 19 snails died on tethers, and 12 snails were crushed/peeled. Daily survival across all sizes was 0.862. The daily cumulative survival for smaller juvenile snail size classes (< 10 mm) was slightly lower (0.821) than survival across all sizes (0.862). Daily survival in predator exclosure cages was almost 100% (cumulative mean = 0.997, se = 0.001, n = 49 days) and did not differ between seasons (overlapping 95% confidence intervals; Figure S4). One of the cages was eliminated from the analysis because it was colonized by a single giant water bug and only empty shells were left by the end of the experiment.

In WCA3A, we observed a total of 276 independent observations of survival across the sites and seasons. After 24 hours, 240 snails survived, 21 snails were left empty, 3 snails had been crushed/peeled, 3 snails died on tethers, and 2 were missing. Only small snails were tethered, and daily survival for these small sizes was higher (0.892) than those in LILA (0.821).

We found the size-dependency of FAS survival changed with seasons. The top four models (cumulative weight = 0.95) for predicting daily survival probability included SL, Season, and the interaction between Length and Season (Table S1). The top model did not include any additional variables, but the next three best models (ΔAIC $\leq$ 2.74) included combinations of spatial factors. The parameter values for the spatial factors appeared to provide little additional predictive capacity (parameter *p-values* $\geq$ 0.276) to survival, so we restricted interpretation to the size and season parameters (Figure S1). During the dry season, FAS daily survival probability increased with size ($z = 2.667$: $p = 0.008$; Figure S1), but in the wet season, daily survival probability was size independent ($z = -0.902$: $p = 0.367$; Figure S1). Small juvenile snails (< 10 mm SL) survived better in the wet season than the dry season (Figure S1).

Table S1: AIC model selection table for logistic regression predicting daily survival probability of FAS (*Pomacea paludosa*) in two LILA wetlands. Daily survival was measured with snails (Length: 3-30 mm SL) on tethers during the dry and wet seasons on transects located closer and further from habitat edges in sloughs.

Model description	AIC	ΔAIC	w
Length + Season + Length*Season	519.870	0.000	0.398
Length + Season + Wetland + Length*Season	520.755	0.885	0.256
Length + Season + Transect + Length*Season	521.482	1.612	0.178
Length + Season + Wetland + Transect + Length*Season	522.387	2.517	0.113
Length + Season	527.249	7.379	0.010
Season + Wetland	527.993	8.123	0.007
Transect + Season + Length	528.705	8.835	0.005
Length + Wetland + Season + Length*Wetland	528.824	8.954	0.005
Transect + Wetland + Season + Length	529.119	9.248	0.004
Season + Wetland + Length + Season*Wetland	529.546	9.676	0.003
Season	529.576	9.706	0.003
Wetland	529.771	9.900	0.003
Transect + Length + Transect*Length	529.844	9.973	0.003
Length	529.982	10.112	0.003
Transect + Season	530.487	10.617	0.002
Transect + Wetland + Season	530.704	10.834	0.002
Length + Wetland	531.284	11.413	0.001
Season + Wetland + Season*Wetland	531.438	11.567	0.001
Transect + Length	531.829	11.959	0.001
Transect + Season + Transect*Season	531.998	12.128	0.001
Length + Wetland + Length*Wetland	532.028	12.158	0.001
Transect + Wetland + Length	533.135	13.265	0.001
Length + Wetland + Season	534.472	14.601	0.000
Transect	535.316	15.446	0.000
Transect + Wetland	535.997	16.127	0.000
Transect + Wetland + Transect*Wetland	537.412	17.542	0.000

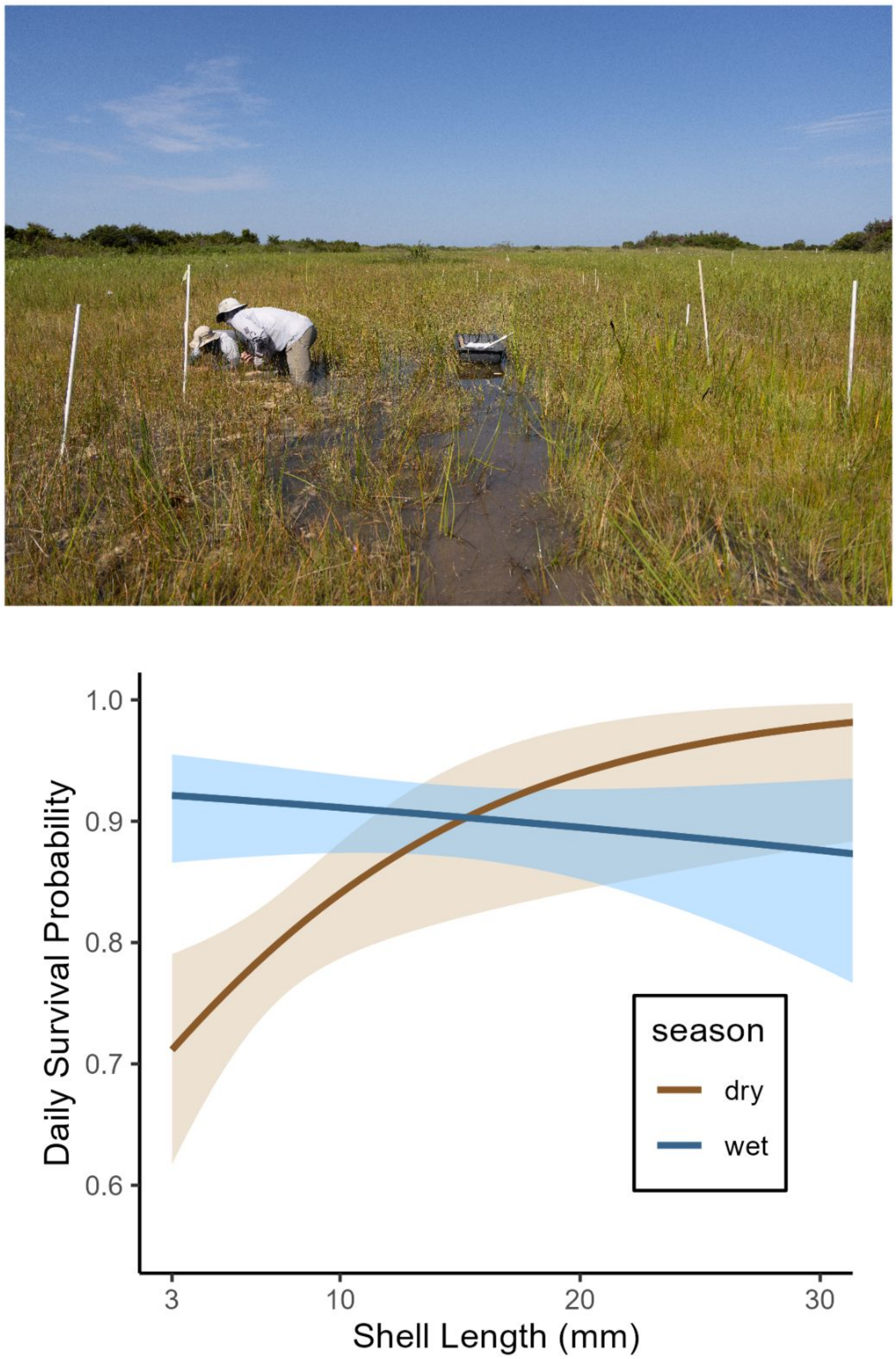

Figure S2: Field picture showing the transects of tethers in LILA wetlands used to estimate daily survival (photo credit: Brandon Güell). Daily survival probabilities estimated from logistic regression from tethering data. Shaded areas indicate standard error.

Predator abundance

Predator abundance was measured using the protocol similar to Dorn & Cook, (2015) (Sommer, 2021a). In both seasons, 1-m^2 throw traps were deployed at 14 randomly selected locations in the slough habitats. Each season sampling occurred when slough habitats were flooded but ridge habitats were shallow (< 10 cm) so for each season large predatory fishes were equally concentrated in the sloughs. Throw traps were cleared under the protocol described by Dorn et al., (2005). Captured animals were euthanized in MS-222 (Tricaine-S, Western Chemical Inc.), fixed (after 30-120 min) in 10% buffered formalin, then cleaned and stored in a 70% ethanol solution. In the lab using calipers, invertebrate predators (i.e., crayfish and giant water bugs) were selected and measured to carapace length and total length, respectively. Juvenile crayfish with carapace lengths < 14 mm were excluded from analyses because they are not predators of juvenile apple snails (Davidson & Dorn, 2017). Trap nets (i.e., fyke and hoop nets) were placed in the deep sloughs of wetlands for three consecutive nights each season. Trapping in each wetland consisted of four fyke nets (0.7 x 1 m opening, 3 mm mesh, 2 throats) and five mini hoop nets (0.6 m diam. opening, 1 cm mesh, 2 throats;) and captures across all gear types were combined to calculate a nightly catch index. Molluscivorous fishes larger than 5 cm were identified, measured (standard length) and released while Greater Sirens were counted and released.

Invertebrate Predator maximum size selection experiment

The purpose of this experiment was to test for the maximum size of apple snail (*P. paludosa*) that a crayfish (*Procambarus fallax*) or giant water bug (*Belostoma lutarium*) would eat. Predators were captured in the Loxahatchee Impoundment Landscape Assessment (LILA) located in Boynton Beach FL using wire minnow traps, then we brought the predators to the green house at the Florida Atlantic University's campus in Davie FL, where they were housed in

1.1 m2 round mesocosms (for crayfish) or 10-gallon aquaria (for giant water bugs). Before placing predators into experimental containers, we measured crayfish and giant water bugs to Carapace Length (CL) and Total Length TL), respectively. Three crayfish, and 5 giant water bugs were then placed into 8 15L-Sterilite containers filled 2/3 full of pond water. In each container, we placed 3 strands of sawgrass (*Cladium jamaicense*; collected from plants already growing at the green house) for giant water bug perching sites, one 3–4-inch piece of 1 inch diameter PVC pipe was added as hiding place for crayfish, and an air bubbler was added in experimental containers to keep the containers well saturated with dissolved oxygen. After starving the predators for 24 hours, we placed a large snail (i.e.,snails larger than the predator could eat; 21-25 mm shell length-SL) into each experimental container for another 24 hours, then we progressively offered a smaller snail (~4 mm SL increments) to each predator for another 24 hours until the predator ate a snail. We measured each snails SL prior to offering the snail to a predator, so we knew the exact SL of each snail that the predator ate. The results of this experiment are summarized in Table S2

Table S2: Table illustrating the results of the predator selection experiment. Each column gives the predator and size while each row shows the SL of snail presented to the predator. The black dots in the cells indicates that a snail in the given size category was presented but not eaten. The cells that contain a number indicate the actual size of the snail eaten by the predator.

	Giant water bug (TL mm)					Crayfish (CL mm)		
Presented *P. paludosa* (SL mm)	22.9	21.9	23.8	22.6	22.0	24.6	27.0	26.6
21-25mm	•	•	•	•	•	•	•	•
17-21mm	•	•	•	•	•	•	•	•
13-17mm	•	13.0	16.3	•	•	•	•	15.3
9-13mm	10.4			10.5	13.5	10.4	9.2	
5-9mm								
3-5mm								

Diet Composition of Mayan Cichlids and Greater Sirens

On the final day of trap netting in the dry and wet season sampling events of 2021, Mayan Cichlids, known to eat freshwater gastropods, were euthanized in MS-222 (Tricaine-S, Western Chemical Inc.), placed on ice, then frozen in the lab for later use in gut-content analysis. Mayan Cichlids and Greater Siren diet samples were analyzed in the lab (gut and fecal samples respectively). The alimentary canal of each Mayan Cichlids was removed and rinsed with 70% ethanol to remove any contents. Greater Siren fecal samples were obtained from Hunter Howell from the University of Miami. The contents were searched and, when possible, identified to lowest possible taxonomic group. The primary goal of the gut content analysis was to find relative sizes of gastropod prey. Whole gastropod in diet samples were measured for Shell Length (SL), but when crushed gastropods were found in diet samples, the apex of the shell was located and compared to apexes of intact shells with known shell lengths.

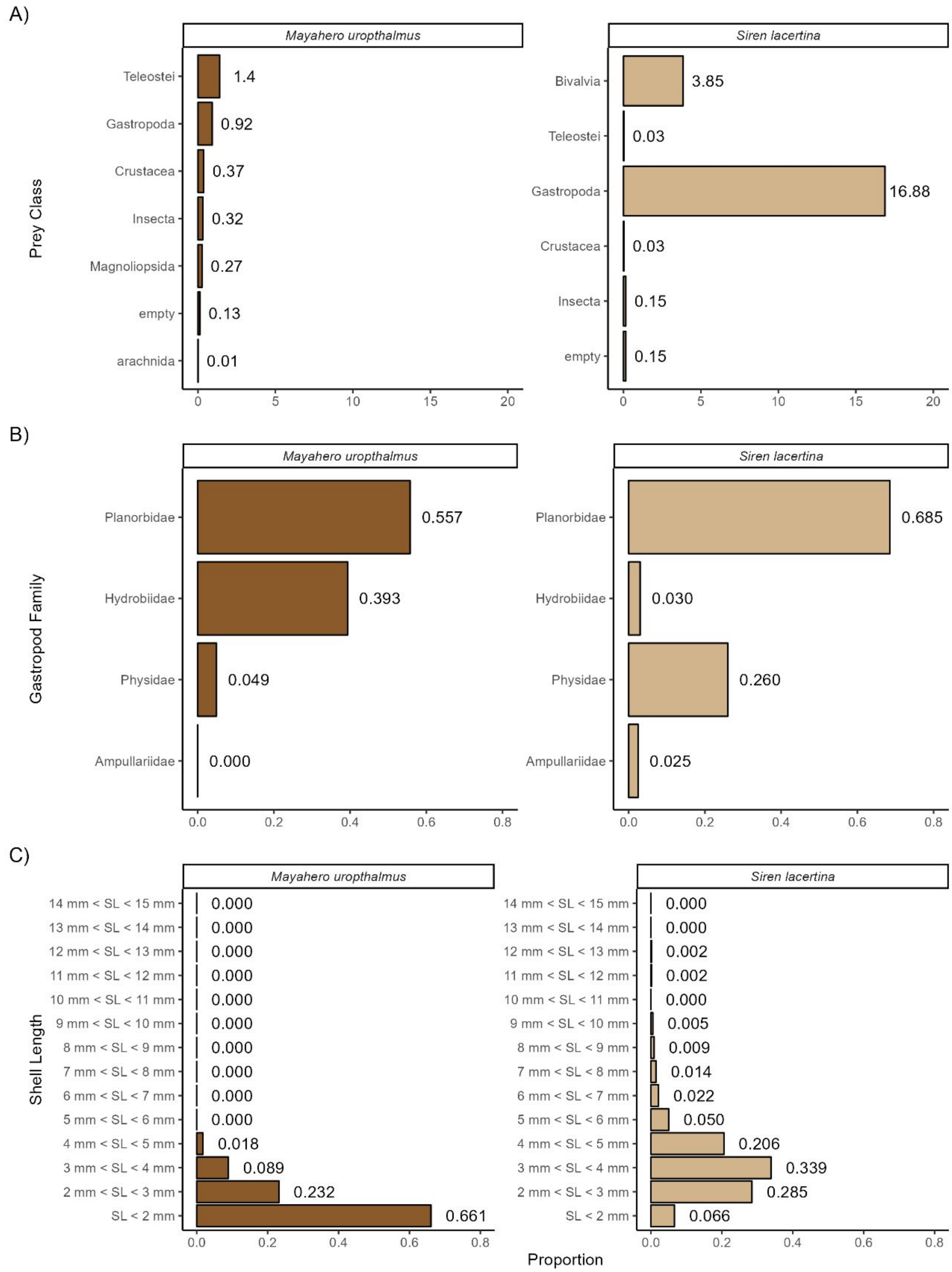

Figure S3: Diet composition of Mayan Cichlids and Greater Sirens. A) per-capita consumption of different orders of prey types. B) composition of gastropod families found within the gastropod portion of the diets. C) size distribution of gastropods found within the gastropod portion of diets.

Relative composition of predation from tethering remains and abundances

For the full tethering experiment in LILA, we determine the relative strength of predation by each juvenile predator between seasons by exploring three different aspects of predation. 1) We looked at the differences in the counts of the three artefacts related to predators (crushed/peeled, empty, missing) across seasons. Crayfish use their mandibles to crush or peel the snail shell to remove the soma (Davidson & Dorn, 2018). In contrast, giant water bugs pierce the snail operculum then suck out and remove snail soma without damaging the shell (Kesler & Munns, 1989). We confirmed the artefactual differences by placing tethered snails in aquarium in the presence of predators. 2) We looked at seasonal changes in abundance of the three predators (i.e., giant water bugs, crayfish, and greater sirens) that were most likely responsible for the artefacts. Predator abundance data was taken from small and large animals sampling in the dry and wet season of 2021 using throw traps and trap nets (i.e., fyke and hoop nets) under a protocol similar to (Dorn & Cook, 2015) (further explained in next subsection and Sommer 2021). 3) We divided the counts of the artefacts by the seasonal abundance of the different predators to estimate per-capita predation rates.

We found tethers retained crushed/peeled shells when consumed by crayfish and empty shells when consumed by giant water bugs Table S2. We interpreted lost snails as vertebrate predation. We examined the stomach and fecal contents of greater sirens and mayan cichlids collected from trap-net monitoring to determine which vertebrate predators was likely to have removed snails from the tethers (Table S2; Figure S2). The size range of snails found in mayan cichlids (snails < 3 mm SL) was typically smaller than hatchling FAS (3 mm SL) whereas the size range of snails found in the diets of greater sirens overlapped the sizes of juvenile FAS (3-10 mm SL; Figure S2). And juvenile FAS were found in the diets of greater sirens but not mayan cichlids (Figure S2). No redear sunfish were caught in the trap nets during this study. From the laboratory, dietary, and capture observations, we interpreted a “crushed/peeled” shell as mortality caused by crayfish (Figure S3), “empty” shell as mortality caused by giant water bugs (Figure S3), a “missing” shell as caused by greater sirens (Figure S3), and “dead” as a caused by something abiotic.

The mortality artefacts of juvenile snails from LILA wetlands (i.e., shell conditions) indicated that there were more than 60% more juvenile predation events in the dry season than the wet season (Figure S3). Giant water bugs, crayfish, and greater sirens were 45, 66, and 77 percent less abundant in the wet season sampling than the dry season, respectively (Figure S3). Except for giant water bugs, per-capita predation (artefacts/abundance) increased in the warmer wet season. Although predator abundance and per-capita predation rates were not explored in WCA3A, the seasonal change in artefact counts in WCA3A were consistent with those found in LILA, except vertebrate predation (missing artefacts) was essentially absent.

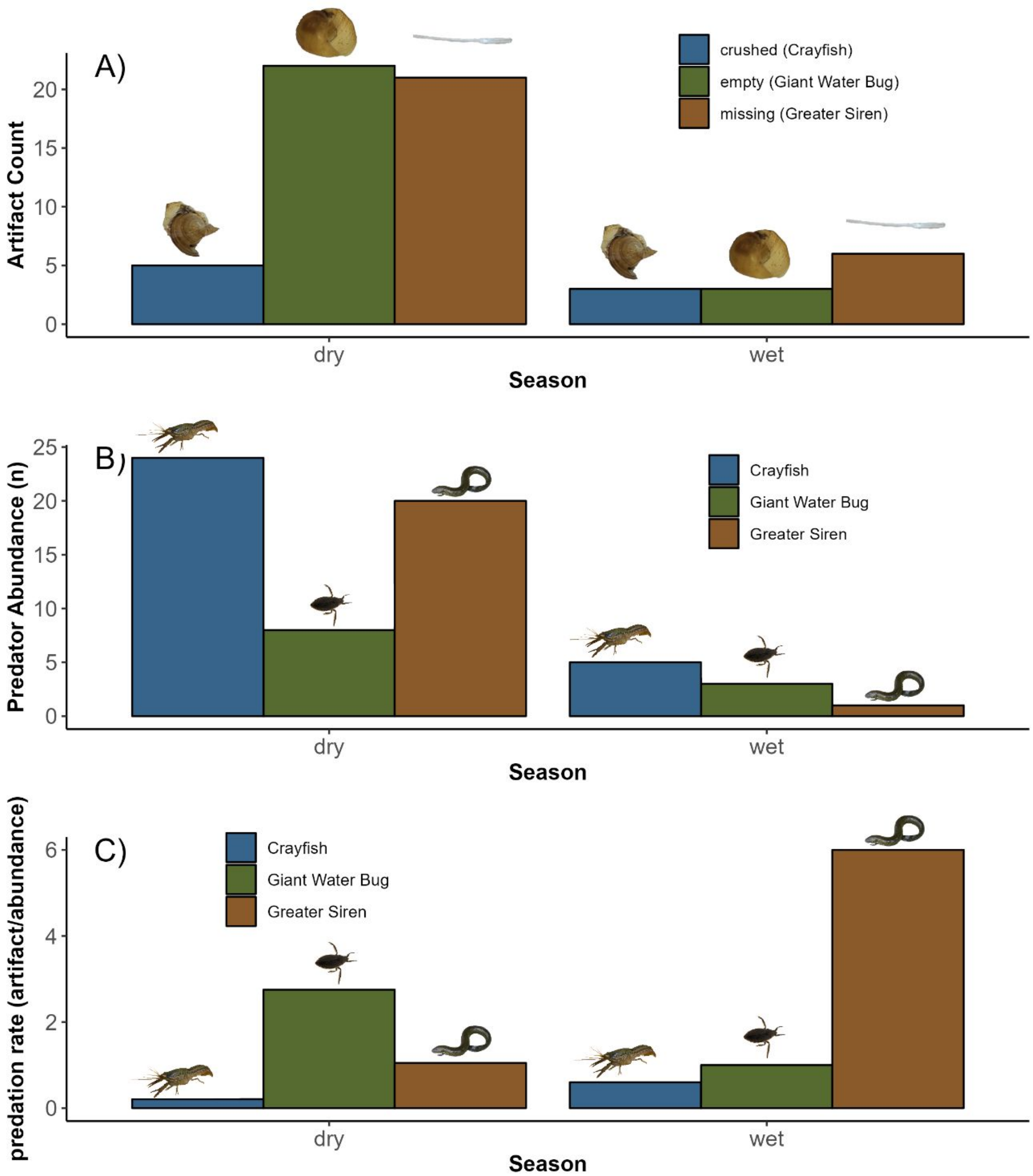

Figure S4: A) Counts of artefacts of biotic factors causing mortality of snails (< 10 mm SL) in the two seasons in the LILA wetlands, and B) seasonal abundance of predators of juvenile snails from throw-trap samples (crayfish and giant water bug), and from standard sets of trap nets (greater siren). Sampling effort was equal in each season. C) Per-capita predation rate from the different predators in the two seasons.

Daily survival in predator exclosure cages

Snail survival was checked at the end of the in situ cage experiment and dead snails (i.e., their shells) were measured for shell length (SL). To obtain the duration that the snail survived I used the modelled growth equation to find the SGR_L that would be expected for that snail's initial size. Using the expected SGR_L for the snail's initial size, the SGR equation can be rearranged to back-calculate the time that the snail lived in the cage:

$$t = \frac{(\log(L_f) - \log(L_i))}{SGR_L}$$

The daily average survival was found by averaging the proportion of snails alive on the given day. If a snail had died on a given day, it was removed from further proportions. One predatory *B. lutarium* colonized a cage in the dry season and all four snails were killed, this cage was excluded from this analysis.

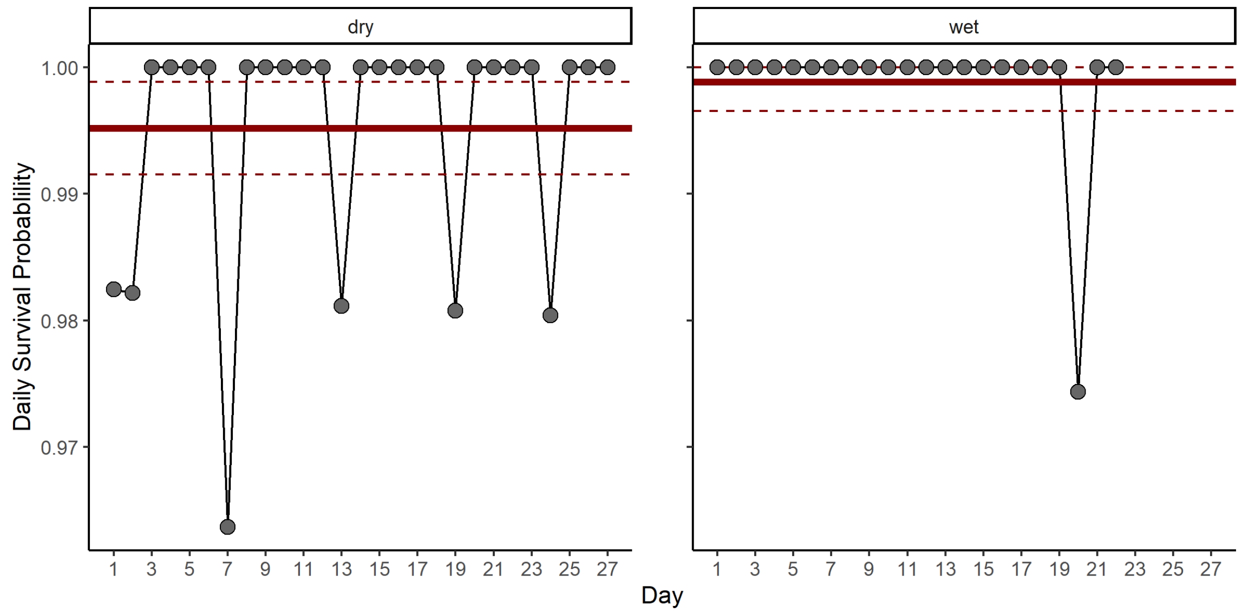

Figure S5: C) The daily survival probabilities obtained from the back calculated time of deaths of snails in the enclosure cages. The solid red line indicates the mean and dashed red lines indicate the 95% confidence intervals for daily survival probabilities across the duration of the experiment.

Calculating the growth parameter

The age-structured population model (Darby et al., 2015) used the following equation to model growth of FAS.

$$Size_{time_i} = \frac{Size_{initial} e^{k_{growth} \cdot time_i}}{1 + (\frac{Size_{initial}}{Size_{max}})(e^{k_{growth} \cdot time_i} - 1)}$$

where time is the duration of growth, and $Size_{initial}$ is the initial length of the snail, $Size_{max}$ is the maximum length that an adult can reach (assumed to be 50 mm SL). Because we knew the $Size_{intial}$, $size_{max}$ and time, we could then calculate k_{growth} for each snail by rearranging the equation.

$$k_{growth} = \frac{\ln\left(size_{time_i}\left(1 - \frac{size_{initial}}{size_{max}}\right)\right) - \ln\left(size_{initial}\left(1 - \frac{size_{time_i}}{size_{max}}\right)\right)}{time}$$

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Journal: Ecology

Article Title: Interpreting field measurements of juvenile growth and survival rates with population growth isoclines

Authors: Nathan T. Barrus, Mark I. Cook & Nathan J. Dorn.

Appendix S3: Environmental Conditions

LILA hydrologic experiment

The water levels in LILA are controlled by pumps and culverts to perform landscape-scale hydrologic experiments. Wetlands M1 & M3 were managed for an unconstrained hydrologic treatment while M2 & M4 were managed for a constrained hydrologic treatment. The unconstrained wetlands are generally deeper than constrained wetlands and depths rise faster in the wet season although wetlands reach the same low water levels in the dry season (Figure S1; below). Shallower water levels are generally favorable for FAS reproduction (Barrus et al., 2023), we refer to the deeper unconstrained hydrologic treatment as "poor reproduction" and the shallower constrained hydrologic as the "good reproduction" hydrologic treatment in the manuscript. The depth flux in LILA are realistic conditions experienced within the natural Everglades landscape but their net effects on population growth are less clear than their impacts on reproduction.

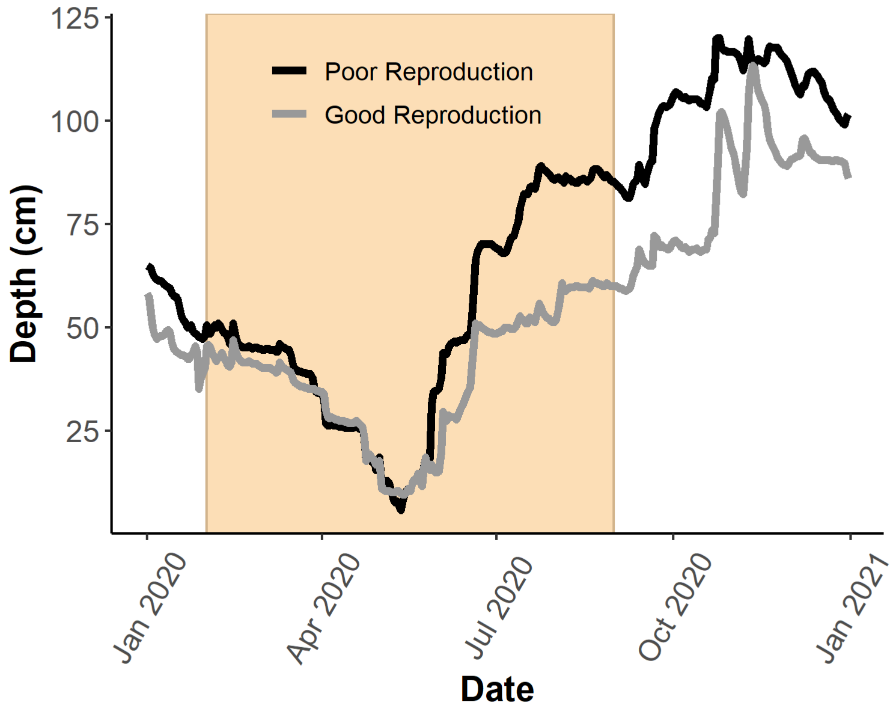

Figure S1: The hydrologic treatments in LILA in 2020. Shaded Area represents the Florida apple snail reproductive season.

WCA3A site level total phosphorus

Table S1: site level differences in periphyton total phosphorus in WCA3A.

Site	Season	$n_{samples}$	Total Phosphorus ($\mu g \cdot g^{-1}$)		
			mean	sd	CI (min,max)
Site 2	wet	2	410.83	21.10	(381.59, 440.07)
Site 3	wet	2	121.94	23.13	(89.89, 153.99)

Seasonal Growth and Temperature

Size-specific growth rates in the wet season (month) were greater than those in the dry season (month, Figure S1). Water temperatures were also warmer in the wet season than in the dry season (Figure S1). Seasonal growth measurements in the WCA wetlands showed qualitatively similar patterns with higher growth in the wet season and lower growth rates in the dry season.

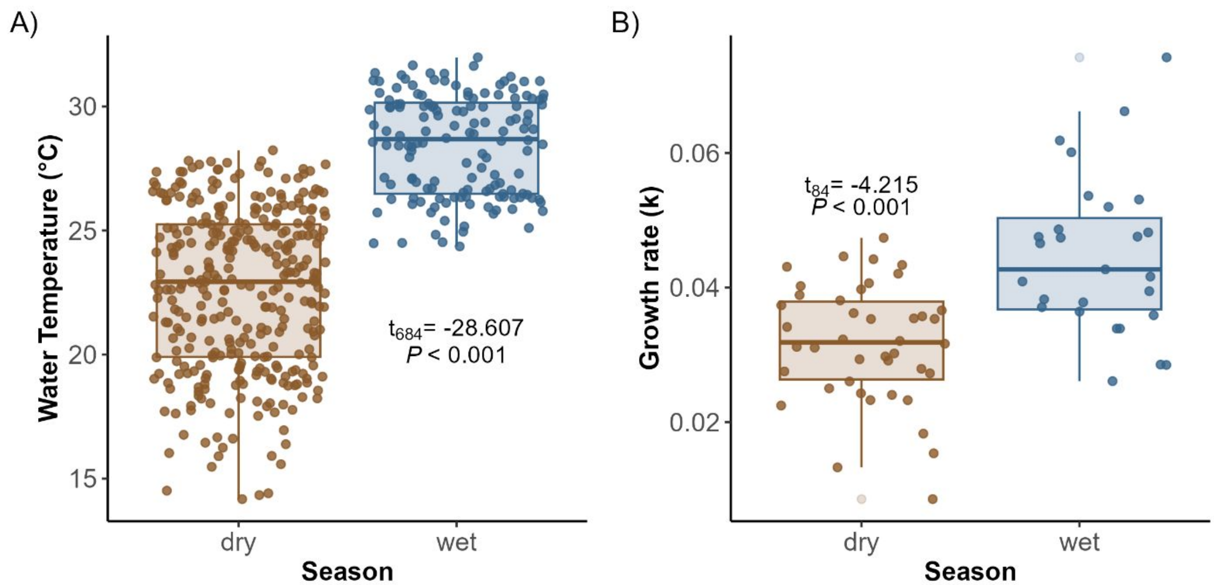

Figure S2: Seasonal A) daily water temperatures and B) Florida apple snail juvenile growth in the LILA wetlands of the Everglades. Each point in panel B represents an individual snail.

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Barrus, N. T., Drumheller, D., Cook, M. I., & Dorn, N. J. (2023). Life history responses of two co-occurring congeneric Apple Snails (Pomacea maculata and P. paludosa) to variation in water depth and metaphyton total phosphorus. *Hydrobiologia*, *850*(4), 841–860. https://doi.org/10.1007/s10750-022-05128-9

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Interpreting field measurements of juvenile growth and survival rates with population growth isoclines

Nathan T. Barrus[1,2,~~4~~3]; ORCID: 0000-0001-7503-3120

Mark I. Cook[~~3~~4],

and Nathan J. Dorn[2,3~~4~~]; ORCID: 0000-0001-5516-0253

[1] Corresponding Author: Nathan T. Barrus, nbarrus1@gmail.com

[2] Current affiliation: Institute of Environment and Dept. of Biological Sciences, Florida International University, Miami, FL, USA

[3] Department of Biological Sciences, Florida Atlantic University, Boca Raton, FL, USA~~Applied Sciences Bureau, South Florida Water Management District. West Palm Beach, FL, USA~~

[4] Applied Sciences Bureau, South Florida Water Management District. West Palm Beach, FL, USA ~~Department of Biological Sciences, Florida Atlantic University, Boca Raton, FL, USA~~

Open Research Statement

~~Data associated with the manuscript will be archived in the Zenodo public repository should the manuscript be accepted. R scripts for analyses are archived in a Git Hub repository ().~~Upon acceptance, all data and code associated with the manuscript will be provided via Zenodo.

Key words

Consumptive effects, size-dependent mortality, demographic rates, interaction strength, predator-prey, top-down vs bottom up.

Abstract:

Juvenile survival and growth rates are commonly studied demographic rates with consequences for population growth. For species that can grow to achieve a size refuge from juvenile predators, the time spent at smaller vulnerable sizes is expected to affect population dynamics, but the ~~combinatorial~~ interactive effects of juvenile growth and survival ~~has not been~~are rarely illustrated theoretically, and most studies of ~~the interaction~~these concepts have been in experimental settings. The ~~combinatorial~~ interactive effects of the two rates have applications to field studies of recruitment variation for a diversity of species that could be assessed with demographic models and isoclines. We conceptually illustrate the potential of using demographic isoclines for marine, terrestrial and freshwater examples in the literature, and then demonstrate the use of a demographic isocline in a case study for an annual freshwater gastropod (Florida Apple Snail, *Pomacea paludosa*). Using a published size-indexed demographic model, we constructed a zero-population growth isocline for theoretical combinations of juvenile growth and survival rates. We then quantified daily juvenile survival and growth in two wetlands twice during the recruitment period, incorporating variable predator assemblages and seasonal environmental conditions (i.e., water depth and temperature). Daily juvenile survival rates were lower in the cooler dry season than in the warmer wet (rainy) season. Juvenile growth was faster in the warmer wet season. Parameter combinations of juvenile growth and survival in the dry season predicted declining populations ($\lambda < 1$), while rates from wet season predicted populations at replacement ($\lambda = 1$) or increasing. When parameters were combined for the full annual recruitment window, populations were projected to decline in both wetlands. The qualitative predictions were robust to variation in hydrologic conditions affecting reproductive rates, but with better hydrologic conditions one lambda was near one (i.e., at replacement). Our

demographic isoclines gave population-dynamic context to field measured demographic rates, identified important temporal variation in survival and growth for the population and generated new hypotheses for future investigation and management. We encourage others to consider developing demographic isocline to interpret variation of stage-sensitive demographic rates across spatiotemporal environmental conditions.

Introduction

Population growth dynamics for many species are influenced by the ~~mediation~~ interaction between individual growth rates and stage- or size-specific mortality rates (Werner and Gilliam 1984, De Roos et al. 2003, Craig et al. 2006). In a review paper 40 years ago, Werner and Gilliam (1984) wrote that size-indexed demographic models are important because a) size is a key feature affecting vital rates, and b) growth rates determine the relationship between size and age. ~~Growth rates are known to mediate survival~~Growth rates and size-dependent survival rates interact to determine the overall number of deaths ~~within a population~~for a cohort ~~rates~~ through at least two mechanisms: 1) higher growth rates during favorable seasons can decrease individual mortality rates through a subsequent stressful season because larger individuals tend to survive stressful conditions better than smaller individuals (Carlson et al. 2008, Viñals-Domingo et al. 2020); and, 2) growth can be a type of defense against stage-specific consumers because growth determines the amount of time an individual spends in vulnerable sizes (Werner and Gilliam 1984, McCoy et al. 2011, Davidson et al. 2021). Growth as a defense against stage-specific consumers has been shown to be important mechanism for populations of amphibians (McCoy et al. 2011), fishes (Rice et al. 1993), terrestrial plants (Brice et al. 2024), as well as terrestrial and aquatic invertebrates (Pepi et al. 2018, Davidson et al. 2021). Research on size-structured interactions has historically focused on theoretical and empirical treatments of density-dependent growth rates, competition, ontogenetic habitat switching, population size-structure, and juvenile bottlenecks (Werner and Gilliam 1984, De Roos et al. 2003). More recent~~ly~~ studies of the interaction have focused on how individual growth rates alone influence cohort survival in size-structured populations (Craig et al. 2006, McCoy et al. 2011, Schmera et al. 2015, Brannelly et al. 2019), ~~but~~ while only two studies have examined the theoretical effects of the interaction~~juvenile growth and mortality~~ between juvenile growth and survival ~~conditions~~ on~~f~~ populations dynamics (e.g., fish recruitment-Rice et al. 1993, equilibrium densities of caterpillars-Pepi et al. 2023). ~~To our knowledge, e~~Explicit theoretical predictions about population growth ~~or declines~~rates are ~~lacking~~ rare and no effort has been made to use theoretical predictions to interpret the combinations of ~~measured~~ field-measured demographic rates on population growth.

Size-indexed demographic models track size at age, combining juvenile growth rates and per-capita survival rates to make population growth projections. Such models can also be used to identify the demographic parameter space making population growth negative, zero, or positive and we suggest the parameter combinations can be profitably illustrated with zero population growth isoclines. Zero population growth isoclines have typically been used theoretically to predict population dynamics and/or coexistence outcomes for interacting species under variable parameter values and assumptions about the interactions (MacArthur and Levins 1964, Vance 1985), but zero-growth isoclines could also be derived numerically from demographic population models to identify parameter combinations producing zero population growth. To our knowledge this has not been done, but demographic isoclines from demographic models ~~that include growth rates~~ could offer quantitative maps for interpreting the ~~combinatorial~~ interactive effects of ~~individual~~juvenile survival and growth of a sensitive early life stage on population growth. Field measured parameters could then be compared to the isocline to identify natural spatial or temporal variation influencing recruitment or population growth. Here we describe examples from marine, terrestrial, and freshwater ecosystems that could~~that can~~ be conceptualized using demographic isoclines and demonstrates the use of an isocline with a case study of a freshwater gastropod of conservation concern.

A demographic~~-based~~ isocline can be created by identifying specific combinations of demographic rate parameters that produce populations at dynamic equilibrium (λ = 1) and plotting the rate combinations in demographic rate space. The two demographic parameters we use in this paper are juvenile survival which we define as the probability an individual of juvenile size will survive from one day to the next, and juvenile growth which we define as the daily change in size or mass~~ of an individual of juvenile size~~. In contrast to juvenile survival which is a daily rate, cohort survival refers to the probability juveniles born in one year survive to adulthood the next year which is the product of the juvenile survival and juvenile growth rate because juvenile growth rates determines the number of days juveniles experience the low daily survival rate in their sensitive stages. For a given life history, the hypothetical isocline in Figure 1A represents where births equal- the juvenile losses (λ = 1) attributed to the product of juvenile growth and survival rates (i.e., cohort survival)-. Populations grow (λ > 1) when ~~a population experiences~~ conditions produce~~ing~~ demographic rates above and to the right of the isocline and decline (λ < 1) when ~~experiencing~~ average rate combinations fall below and to the left. The exact shape (i.e., steepness, linearity) of the isocline will depend on life history characteristics like longevity, adult survival, and fertility/reproductive rates that could be mediated by the environment or density-dependence. The stability of the isocline would depend on the relative constancy of the other life history characteristics of the ~~species~~population. If juveniles can escape

Commented [ND1]: A cohort = all the individuals born at the same time (e.g., year). Does this definition still work?

When you write "juveniles" six words later, do you mean all juveniles (a cohort)? Or do you mean probability one juvenile survives to adulthood? Seems like the latter.

vulnerable sizes through faster growth rates, then a negative slope would be expected (Figure 1A). In natural settings, spatiotemporal environmental factors that influence juvenile survival and growth will combine to mediate where the population ~~falls~~exists in demographic rate space (Craig et al. 2006, Davidson and Dorn 2018, Davidson et al. 2021, Ma et al. 2021, Nunes et al. 2021, Meehan et al. 2022). We emphasize that the isocline is theoretical and identifies parameter values producing λ=1 from a size-indexed demographic model with other life stage parameters held constant. Density dependence in a real population could act at the juvenile stage and produce rates ~~falling~~ on the isocline, but should not fundamentally change the isocline. In contrast, if~~f~~ density-dependence acts on the adult stages it would conceivably affect the isocline (i.e. ~~chang~~altering reproductive rates) and we discuss this in our case study, but do not explore the details further here. If most of the variation in juvenile survival can be attributed to predators (Werner and Gilliam 1984) then the ~~rate~~ locations of the rates in demographic state space indicate the degree of consumer control on the population and environmental mediation of such top-down control (Figure 1B-D).

~~A demographic-based isocline can be created by identifying specific combinations of demographic rate parameters that produce populations at dynamic equilibrium (λ = 1) and plotting the rate combinations in demographic rate space. The two demographic parameters we use in this paper are juvenile survival which we define as the probability an individual of juvenile size will survive from one day to the next, and juvenile growth which we define as the daily change in size or mass of an individual of juvenile size.In contrast to juvenile survival which is a daily rate, cohort survival refers to the probability juveniles survive to adulthood which is the product of the juvenile survivaland juvenile growth rate because juvenile growth rates determine the number of days juveniles experience the daily survival rate. For a given life~~

~~history, the hypothetical isocline in Figure 1A represents where births equal the juvenile losses (λ = 1) attributed to the product of juvenile growth and survival rates (i.e., cohort survival) . Populations grow (λ > 1) when a population experiences conditions producing demographic rates above and to the right of the isocline and decline (λ < 1) when experiencing average rate combinations below and to the left. The exact shape (i.e., steepness, linearity) of the isocline will depend on life history characteristics like longevity, adult survival, and fertility/reproductive rates that could be mediated by the environment or density-dependence. The stability of the isocline would depend on the relative constancy of the other life history characteristics of the species. If juveniles can escape vulnerable sizes through faster growth rates, then a negative slope would be expected (Figure 1A). In natural settings, spatiotemporal environmental factors that influence juvenile survival and growth will combine to mediate where the population falls in demographic rate space (Craig et al. 2006, Davidson and Dorn 2018, Davidson et al. 2021, Ma et al. 2021, Nunes et al. 2021, Meehan et al. 2022). We emphasize that the isocline is theoretical and identifies parameter values producing λ=1 from size-indexed demographic model with other life stage parameters held constant. Density dependence in a real population could act at the juvenile stage and produce rates falling on the isocline, but should not fundamentally change the isocline. If density-dependence acts on the adult stages it would conceivably affect the isocline (changing reproductive rates) and we discuss this in ou case study, but do not explore the details here. If most of the variation in juvenile survival can be attributed to predators (Werner and Gilliam 1984) then the rate locations in demographic state space indicate the degree of consumer control on the population and environmental mediation of such top-down control (Figure 1B-D).~~

~~The isocline approach could be of potential general interest, particularly in its application to populations with high recruitment variation. The potential utility of demographic isoclines is~~

~~demonstrated here by depicting hypothetical isoclines for three taxa of conservation or management concern that span terrestrial, marine, and freshwater ecosystems where the details of recruitment have been sufficiently described (Figure 1B-D). In all three cases, consumption is the primary mechanism driving survival of juvenile stages, and the demographic parameters related to environmental gradients have been well studied. Thus,the qualitative predictions about where conditions fall relative to the isocline can be reasonably inferred (Figures 1B-D).~~ The isocline approach could be of potential general interest, particularly in its application to populations with high recruitment variation. The potential utility of demographic isoclines is demonstrated here by depicting hypothetical isoclines for three taxa of conservation or management concern that span terrestrial, marine, and freshwater ecosystems where the details of recruitment have been sufficiently described (Figure 1B-D). In all three cases, consumption is the primary mechanism driving survival of individuals in juvenile stages, and the demographic parameters related to environmental gradients have been well studied. Thus, the qualitative predictions about where conditions fall relative to the isocline can be reasonably inferred (Figures 1B-D). By describing these three examples from the published literature under the same framework we do not intend to over-simplify the ecological details of recruitment, but rather to conceptually illustrate the similar issue of the demographic rate combinations that can help population biologists, whether involved in conservation, resource management or pest management, assess the potential for population growth using a model and field-measured rates.

First, in a freshwater rock pool system a spatial temperature gradient simultaneously affects growth rates of larval mosquitoes (*Aedes atropalpus*) and their survival with dragonfly (*Pantala* spp.) naiad predators (i.e., per-capita foraging rate of the predator increases with temperature; Figure 1B; Davidson et al. 2021, 2024). The net effect of both rate changes was such that at cooler temperatures the mosquito populations would not recruit well even though survival was high because daily growth was too slow; rate combinations ~~were~~ would be below and left of the isocline (Figure 1B). With higher temperatures the average daily survival decreased, but the increased daily growth rate with warmer temperatures shortened the time spent in the larval stage even further so that mosquito populations could recruit and grow; joint rates moved down, but also to the right of the isocline (Figure 1B). In a contrasting invertebrate system (not shown) of ant predator-caterpillar prey interactions in a terrestrial system, warmer temperatures affected both rates in the same manner, but increases in caterpillar growth with higher temperatures were unable to compensate for lower survival from ant predation (Pepi et al. 2018).

In a well-studied terrestrial ecosystem, recruitment (sucker to sapling transition) of quaking aspen (*Populus tremuloides*) that are browsed by elk (*Cervus canadensis*) are influenced by elk numbers and environmental variation (Figure 1C; Brice et al. 2024). After wolves (*Canus lupis*) were extirpated from the Greater Yellowstone Ecosystem, USA (GYE) and elk abundance was high, the browsing pressure on aspen stands was high everywhere regardless of the available moisture, resulting in widespread recruitment failure (bottom of Figure 1C). Following wolf reintroduction the elk declined and browsing pressure was reduced (Kauffman et al. 2010, Brice et al. 2024), but aspen stand regrowth was variable (Ripple and Beschta 2007, Kauffman et al. 2010, Beschta and Ripple 2016). Recent evidence indicates that the patchy recruitment of aspen suckers from different stands may be from spatiotemporal variability in moisture allowing aspen suckers to grow to sizes large enough to escape the browsing pressure (upper points in Figure 1C; Brice et al. 2024). While no comparison of browsing rates and sucker growth has been conducted to our knowledge, this understanding of spatially patchy aspen recruitment in GYE was proposed by Kauffman et al. (2010) and the work by Brice et al. (2024) confirmed~~s~~ the importance of the interaction from the aspect of spatiotemporal variation in moisture. Examinations of the interactive effects of the two rates ~~in combination~~ across spatiotemporal gradients, rather than being considered as alternative explanations (i.e., top-down vs. bottom-up factors), could improve the understanding of the patchy regrowth of aspen and other plants in response to predator reintroductions and herbivore densities (Beschta and Ripple 2016).

In a marine ecosystem, multiple environmental gradients (e.g., depth, salinity; Munroe et al. 2017, Baillie and Grabowski 2019) appear to affect one or both demographic rates of small settling oysters (e.g., *Crassostrea virginica* or *Saccostrea glomerata*) which are considered foundation species. Higher salinity produces faster growth of small oysters (Munroe et al. 2017), but high salinity can also facilitate outbreaks of predaceous drilling snails (Kimbro et al. 2013). How the two rates change together in state space with increased salinity is unclear from the literature, but salinity-mediated changes to survival and a subsequent population collapse (illustrated in Figure 1D) ~~in an estuary~~ was observed by Kimbro et al. (2013) in an estuary.

While the specific slopes of the isoclines for these examples from freshwater, marine and terrestrial ecosystems are unknown, the qualitative description of the isocline should be generalizable (i.e., negative slope) to any species with high mortality from stage- or size-dependent juvenile predators. As a case study, we examined an annual gastropod of conservation concern, the Florida Apple Snail (*Pomacea paludosa;* hereafter FAS), to demonstrate the utility of a demographic~~-based zero population growth~~ isocline. We identify theoretical combinations of juvenile-stage parameters predicting population stasis ($\lambda = 1$), growth ($\lambda > 1$), or decline ($\lambda < 1$) from a previously parameterized size-indexed demographic~~age-structured~~ model. Then, we quantified daily size- and season-dependent survival and juvenile growth rates in the field to interpret the combined effects that juvenile growth and survival have on predicted population growth ~~during the annual recruitment period~~. Using the isocline approach, the field measured values were interpretable from a population-dynamic perspective and provided material for generating novel hypotheses about population limitation.

Materials and methods

System and study species

The Florida Everglades is a shallow, expansive (~915,000 ha), subtropical, oligotrophic wetland covering much of southern Florida (Richardson 2010; Appendix S2: Figure S1). Rainfall is seasonal with approximately 80% of rain falling in the wet season from June–November (Gaiser et al. 2012). which produces intra-annual water depth fluctuations of ≥ 60 cm. The degree of water level recession and depth in the dry season is a function of rainfall and water management decisions. Historically, water flowed in a single shallow sheet from Lake Okeechobee at slow velocity across the spatial extent of the Everglades (i.e., sheet flow; Sklar et al. 2005), but flow was reduced or eliminated after compartmentalization and drainage. Drainage of the Everglades altered the hydrologic conditions by increasing water depths in some areas but decreasing depths in others. Within the Everglades, the ridge-slough landscape originally covered 55% of the Everglades (McVoy et al. 2011), but now covers ~44% (Richardson 2010). In the ridge-slough landscape, ridges and sloughs differ slightly by elevation (~10–15 cm) which produces habitat/vegetation patterning. The lowest elevation slough habitats dry to sediment surfaces every 3–10 years and are dominated by floating vegetation like water lilies (*Nymphaea odorata*) or emergent spike-rushes (*Eleocharis* spp.). Sloughs are interspersed with higher elevation ridges dominated by sawgrass (*Cladium jamaicense*) that dry most years (Zweig and

Kitchens 2008). Ongoing hydro-restoration of the Everglades ecosystem aims to restore hydro-patterns to improve conditions for wildlife and natural communities.

The FAS is the largest native freshwater gastropod in North America, it inhabits shallow lakes and wetlands, and currently occurs at low adult densities (< 1/m^2) in southern Florida wetlands (Gutierre et al. 2019). Snails grow from 3–4 mm shell length (SL) at hatching to > 40 mm SL as adults and do not live beyond 1.5 years (Hanning 1979). Most reproduction (~70%) occurs during cooler spring seasons when water levels are declining, and some reproduction occurs (~30%) during warmer early summer when water levels are rising (Hanning 1979, Barrus et al. 2023). At adult sizes (> 25 mm SL) FAS are a critical resource for the endangered Snail Kite (*Rostrhamus sociabilis*; Cattau et al. 2014), which experienced significant declines within the ridge-slough landscape from 2001–2010. The decline in Snail Kite populations in the Everglades is at least partly explained by declines in FAS, so improving the conditions within the ridge-slough landscape for FAS populations is imperative. As small juveniles (< 10 mm SL) FAS are prey for crayfish (*Procambarus* spp), sunfish, non-native cichlids, large killifishes (*Fundulus seminolis*), greater siren (*Siren lacertina*), and turtles (e.g., *Kinosternon bauri*; Valentine-Darby et al. 2015, Davidson and Dorn 2017). Giant water bugs (Belostomatidae, Kesler and Munns 1989) live in the Everglades, are known to eat snails but ~~had~~ have not been studied. Juvenile FAS outgrow most common fish and invertebrate predators when they reach ~10–11 mm SL (Valentine-Darby et al. 2015, Davidson and Dorn 2017, Appendix 2: Figure S2). There has been no investigation of the factors that influence survival of juvenile snails after hatching or the population-dynamic impacts of cohort survival within natural ~~systems~~ wetlands ~~partly~~ mostly because tracking cohorts of small juvenile snails is logistically infeasible, thus we developed an isocline approach to investigate spatiotemporal variation in ~~mortality and~~ juvenile growth and survival using a demographic model.

Zero-Population Growth Isocline

~~We used a published size-indexed population model (Darby et al. 2015) to create zero-population growth isoclines from theoretical combinations of two parameters, juvenile growth and survival (FAS < 10 mm SL) holding all other variables stable (details in Appendix S1). The model is an age-structured matrix model (i.e., Leslie Matrix) that tracks annual cohorts on daily time steps, but model structure provides the index for size using an equation that describes size at age. Survival and fecundity rates are determined by the size-structure. Cohorts are produced through fecundity rates that are a function of reproductive season (i.e, spring or summer) and water depths. The model was originally developed to provide ecological assessment of hydrologic restoration (tied to water depths) in the Everglades for FAS, butwe re-coded the model in R. We used most of the original parameters from Darby et al. (2015) but excluded carrying capacity and included a few adjusted parameters to reflect recent changes in the understanding of FAS life history (see Appendix S1: Table S1). Numerical simulations were used to construct zero population growth isoclines from combinations of the parameters for juvenile survival and growth under two different hydrologic conditions which produced depth-dependent differences in reproduction (i.e., "Good Reproduction" or "Poor Reproduction"; Appendix S1). For each simulation, we used one juvenile growth rate, one juvenile survival rate, and one hydrologic condition, then we allowed the population to reach a stable size distribution. After a stable size distribution was achieved, we calculated the population growth rate (λ).. After repeating this for 1666 different juvenile growth, mortality and hydrologic conditions, we identified the parameters where population growth was constant (λ = 1). We plotted these values~~

~~as an isocline for each hydrologic scenario (For further details see Appendix S1).~~ We used a published size-indexed demographic population model (Darby et al. 2015) to create zero-population growth isoclines from theoretical combinations of two parameters, juvenile growth and survival (FAS < 10 mm SL) holding all other variables stable (details in Appendix S1). The model is an age-structured matrix model (i.e., Leslie Matrix) that tracks annual cohorts on daily time steps, but the model structure also~~provides the~~tracks size on a daily step~~ index for size~~ ~~using~~with an equation that describes size at age. Survival and fecundity rates are determined by the size-structure. Cohorts of juveniles are produced during the reproductive season (i.e., spring or summer) as a function of fecundity rates and water depths (i.e., depth dependent egg laying rates). The model was originally developed to provide ecological assessments of hydrologic restoration ~~(tied to water depths) in the Everglades for FAS,~~ but we re-coded the model ~~in~~using R for our research application. We used most of the original parameters from Darby et al. (2015), but excluded the carrying capacity and included a few adjusted parameters to reflect recent changes in the understanding of FAS life history (see Appendix S1: Table S1). Numerical simulations were used to construct zero population growth isoclines by adjusting ~~from combinations of the parameters for~~ juvenile survival and growth parameters under two different hydrologic conditions which produced depth-dependent differences in reproduction (i.e., "Good Reproduction" or "Poor Reproduction"; Appendix S1). For each simulation, we used one juvenile growth rate, one juvenile survival rate, and one hydrologic condition, then we allowed the population to reach a stable size distribution. After a stable size distribution was achieved, we calculated the population growth rate (λ). After repeating this for 1666 different juvenile growth, mortality and hydrologic conditions, we identified the parameter combinations ~~s where population growth was constant~~~~producing~~that produced static population sizes ~~(~~λ = 1~~)~~. We plotted these values as an isocline for each hydrologic scenario (For further details see Appendix S1).

The isoclines graphically represent theoretical combinations of the two parameters that stop ~~growth of the~~ population growth ($\lambda = 1$). The isoclines are boundary conditions between a growing or a declining population assuming 1) a static hydrologic scenario and adult life history, and 2) that the ~~given~~ juvenile growth and survival rates represent ~~an~~ the average rate experienced by juvenile snails ~~throughout a year~~for the year. Because juvenile FAS densities are so low in our study wetlands (typically << $0.1/m^2$) and yet juveniles can survive and grow to high sub-adult densities ($16/m^2$) in mesh predator-exclusion cages (Barrus et al. 2023), we considered negative density-dependent growth to be irrelevant to ~~our model evaluations which were trying to identify parameters that would produce an increasing or decreasing population~~the rates the populations of FAS actually experience.

~~Once the isocline was~~After constructing isoclines~~ed~~, we used *in situ* experimental techniques ~~(detailed further below and in Appendix S2)~~ to calculate~~d~~ juvenile survival and growth parameters and their 95% confidence intervals, ~~then we plotted on the~~and plotted them in ~~isocline~~ demographic state space ~~the combination of juvenile survival and growth for each season and location~~. The nature of the model made it impossible to change juvenile growth rates seasonally so~~; thus,~~ the ~~predictions~~ interpretations of the rates in~~from the~~ ~~isocline~~ state space assumed that the parameters were averages experienced throughout the year. The season-dependent predictions ~~from the field measures were therefore expected snail recruitment assuming the rates measured each season~~from the field measurements were, therefore, the expected snail recruitment given the rates measured for the respective season.~~.~~ Because ~70% of reproduction (egg laying) occurs in the dry season and ~30% occurs in the wet season (Darby et al. 2015, Barrus et al. 2023), we calculated the weighted averages of the seasonal parameters to obtain annual average values.

Survival and Growth in the field

We measured juvenile survival and growth in wetlands at the Loxahatchee Impoundment Landscape Assessment (LILA) and in two sites in the western portion of Water Conservation Area 3A (WCA3A; Appendix S2: Figure S1) in Florida, USA. LILA consists of four 8 ha impounded wetlands with ridge and slough elevation features and hydro-patterns that mimic the wetlands of the Everglades (Appendix S2: Figure S1B). Water levels vary seasonally in the wetlands but the water levels in LILA were controlled by pumps and culverts to perform a landscape-scale hydrologic experiment. We worked in two wetland impoundments at LILA that had hydrologic conditions deemed good for FAS reproduction (Barrus et al. 2023). The two sites near the western boundary of WCA3A near Big Cypress National Park (Appendix S2: Figure S1: Sites 2 and 3 in Ruetz et al. 2005) were within a 1240 km^2 contiguous portion of the Everglades. The WCA3A sites were chosen because they were near locations of higher FAS densities in the recent past, including sites that previously supported Snail Kite nesting (Cattau et al. 2014).

We tethered snails by attaching monofilament to the apex of the shell using super glue, then attaching the other end of the monofilament to PVC poles within the wetland (Appendix S2). Tethered snails were placed on transects in the wetlands ~2 m apart and checked daily. Surviving snails were moved to increase independence between nights while depredated snails were replaced. Although in LILA we tethered snails of all sizes to test for size-dependent, here we focus on the survival of snails < 10 mm SL because this related to the isocline and was what varied the most seasonally (Appendix S2). We only tethered snails < 10 mm SL in the WCA3A. Further details of the tethering experiment can be found in Appendix S2. Here we also focus on relating survival to the isocline, but we also observed tethering artefacts of different predators that allowed us to identify common predators (see further discussion in Appendix S2).

Tethering can inflate mortality rates of animals capable of escape (Baker and Waltham 2020), but FAS are relatively less mobile than their typical predators and rely on retracting into their shell to avoid predation rather than escape. For prey with little escape capability, tethering should give reliable information about prey survival particularly across gradients of predation as it measures encounter rate variation (Rochette and Dill 2000, Ruehl and Trexler 2013). Prior experimental work with this species did not find any measurable anti-predator response, either morphological or behavioral, to chronic exposure to crayfish (Davidson and Dorn 2017).

We measured juvenile growth either using *in-situ* 1-m^2 mesh cages or with a regression that predicted wet season juvenile snail growth using total phosphorus (TP) concentrations in metaphytic mats ($R^2 = 0.85$; Barrus et al., 2023). The metaphyton (sometimes called periphyton) in the Everglades is a floating calcareous mat composed of algae, cyanobacteria, other microbes, and algal detritus (Gaiser et al. 2011). Algae was allowed to accumulate in the cages two weeks prior to the experiment, and two liters of metaphyton was placed inside the cages as a food source (Drumheller et al. 2022, Barrus et al. 2023). Juvenile snails were individually marked and placed in cages to grow for 4–5 weeks. We placed 8 cages in LILA during both seasons and 3 cages in WCA3A site 2 in the dry season. To estimate wet season juvenile growth at WCA3A site 3, we measured the TP of metaphytic mats to predict FAS growth using regressions from (Barrus et al. 2023). We were only able to obtain dry season growth rates for site 2 in WCA3A because low dry season water depths at site 3 made use of cage mesocosms impossible. Using the growth results we then calculated the growth parameter k_{growth} to relate the results to the isocline. K_{growth} is a measure of size-dependent daily growth rates that can be calculated from knowing the initial size, the final size and the maximum size. The maximum size was assumed to be 50 mm SL. Details on calculating the growth parameter (k_{growth}) can be found in Appendix S2.

Results

Zero-population growth isoclines created from the size-indexed age-structured population model produced a declining isocline consistent with the expected interaction between daily juvenile growth and survival rates (Figure 2). Combinations of the two parameters above and to the right of the isocline predict growing populations (λ > 1) while combinations below the isocline predict declining populations (λ < 1). The ~~shape~~ negative slope of the isocline illustrates that ~~environmental~~ populations ~~experiencing faster~~with faster juvenile growth ~~conditions~~ ~~favorable producingfor faster juvenile growth will support populations that~~ can withstand ~~higher rates of mortality (i.e., lower daily survival)~~lower daily survival. ~~and p~~Populations ~~with experiencing~~ slower juvenile growth rates ~~growing juveniles~~ ~~can only persist or grow (λ≥ 1) with~~will need higher juvenile survival rates to persist or grow (λ≥1). ~~need~~ ~~lower mortality (i.e, higher daily survival) to persist or grow .(λ≥ 1).~~ Hydrologic conditions that improved reproductive rates~~conditions~~ (i.e., eggs laid/female) moved the isocline down and to the left (gray isocline in Figure 2), making the population slightly more resilient to lower juvenile survival (e.g., withstanding 3.1% lower survival at juvenile growth of k_{growth} = 0.07) and/or slower juvenile growth (e.g., withstanding ~~by~~ 7.7% slower juvenile growth at juvenile survival rates of 0.80; Figure 2). The separation between the isoclines was greatest for ~~conditions~~ combinations of~~with~~ faster juvenile growth and lower juvenile survival (Figure 2).

Empirical Juvenile Survival and Growth related to the Isocline

We observed variation in the measured juvenile survival and growth parameters across sites and seasons (Figure 3). Tethering snails from hatchling to adult sizes indicated that survival was strongly size-dependent in the dry season with snails < 10 mm SL heavily depredated (Appendix S2: Figure S2,~~ Appendix S2:~~ Table S1). ~~Examination of t~~The shell artefacts of deceased snails (< 10 mm SL) suggested that the predators were primarily native invertebrates, ~~(~~*Belostoma lutarium* and~~,~~ *Procambarus fallax*, plus~~)~~ ~~and~~ salamanders (Appendix S2). ~~and~~ ~~p~~Predator surveys indicated that abundances were also variable ~~across~~ between seasons and sites (Appendix S2).

Across both field sites the juvenile growth was faster in the warmer wet season than the dry season (Figure 2, Appendix S3: Figure S2). The dry season had lower juvenile survival and slower juvenile growth with combinations falling below and to the left of the isocline (Figure 2). In contrast, the wet season had higher juvenile survival ~~rates~~ and faster growth; with average combinations falling on the isocline (LILA wetlands) or even above and to the right (WCA3A site 2; Figure 2). Snails in WCA3A site 2 had faster juvenile growth than those in LILA (Figure 2). The combined effects, weighted by seasonal egg ~~production~~ laying distributions, resulted in average ~~mortality and growth~~juvenile survival and growth parameters that predicted a declining population for both ~~sites~~wetlands, though confidence intervals slightly overlapped the zero-population growth isocline for WCA3 site 2 (Figure 2). The overlap of the confidence ~~region~~ interval with the isocline, ~~(~~indicating potential ~~replacement)~~population stasis, could only be observed for an~~when the~~ isocline reflecting~~ed~~ good hydrologic conditions for egg-laying in WCA3A site 2 (Figure 2).

Discussion

Using a size-indexed age-structured population model we produced zero-population growth isoclines illustrating the ~~combinatorial~~ interactive effects of juvenile growth and survival of a predator-s~~s~~ensitive juvenile stage on population growth. The expected effect that faster juvenile growth can offset higher mortality was illustrated. Our work was specific to an annual freshwater gastropod with size-dependent survival, but the approach is conceivably applicable to any size-structured consumer-resource interaction. For example, field combinations of both demographic rates for a marine shrimp were examined for multiple populations by Chockley et al. (2008), but they were not compared against population dynamic predictions. The isocline from the demographic model allowed us to interpret field measured rates and conclude that the populations should be static or declining. Seasonal parameters further indicated that both juvenile survival and growth were poorer in the dry season (spring: Feb–April) which overlaps with most of the reproductive period of FAS. The results produce novel hypotheses about environmental variation and predator control that might limit the FAS in the Everglades. Creating similar demographic isoclines for other species could offer insights into the spatiotemporal conditions producing population growth and recruitment.

Growth-Survival Isocline

~~Our zero-population growth isocline was created using a specific model for an annual gastropod of conservation concern which presents some limitations to the generality. Future theoretical work with a general rather than a specific size-indexed demographic model, should be explored to determine how other parameters might influence the shape of the demographic isocline. Because current densities of FAS are extremely small compared to what the environment can support when predators are excluded, we did not consider negative density dependence to be important for identifying conditions that would support a growing population. While density dependence influences population dynamics acting on juvenile demographic rates (Accolla et al. 2024), it is important to remember that our demographic isocline represents where births equal the juvenile losses ($\lambda = 1$) attributed to the product of juvenile growth and survival rates (i.e.,~~

~~cohort survival) when all other life history parameters are held constant.. Thus, density-dependence acting on juvenile growth and survival should not affect the shape of a theoretical isocline, but density-dependence or other factors (e.g., environmental variation) that affect birth rates would alter the shape of the isocline. Our exploration for the FAS demonstrated that environmental factors (water depth variation) influenced reproduction by altering the slope of the isocline (Figure 2). Exploring the factors that change the shape of the isocline would be a fruitful area for general exploration. While we built our theoretical isocline assuming density-independent population growth we emphasize that this does not assume the actual rates, or an actual population are density-independent. Rather, density-dependence acting on one or both juvenile rates could be an explanation for where the rates exist in state-space. It is the job of the investigator to determine what drives variation in the two rates.~~ Our zero-population growth isocline was created using a specific model for an annual gastropod of conservation concern which presents some limitations to the generality. Future theoretical work with a general ~~rather than a specific~~ size-indexed demographic models, should be explored to determine how other parameters might influence the shape of the demographic isocline.~~–~~ ~~Because current densities of FAS are extremely small compared to what the environment can support when predators are excluded, we did not consider negative density dependence to be important for identifying conditions that would support a growing population.~~ ~~–~~While density dependence influences population dynamics- acting on juvenile demographic rates (Accolla et al. 2024), it is important to remember that our demographic isocline is theoretical and represents rates where births equal the juvenile losses (λ = 1) attributed to the product of juvenile growth and survival rates (i.e., cohort survival) ~~when~~assuming all other life history parameters in a matrix are held constant~~-~~. Thus, density-dependence acting on juvenile growth and survival of real populations should not affect the shape of a theoretical isocline, but only the actual empirical juvenile rates observed. In contrast, ~~but~~ density-dependence or ~~other~~environmental factors ~~(e.g., environmental variation)~~ ~~that~~ affecting birth rates (e.g., adult fecundity or longevity) would ~~alter~~require a different ~~the shape of the~~ isocline as populations grow or decline. Our exploration for the FAS demonstrated that ~~environmental factors (~~water depth variation that~~)~~ influenced ~~reproduction by~~egg laying alter~~ing~~ed the slope of the isocline (Figure 2). Exploring the factors that change the shape of demographic ~~the~~ isoclines would be a fruitful area for more general exploration. –While we built our theoretical isocline assuming density-independent population growth we emphasize that this does not assume the actual juvenile rates~~, or an actual population~~ are density-independent. Our isocline does not claim anything about the relative importance of density dependence on population growth. Rather, density-dependence acting on one or both juvenile rates could be an explanation for where the rates exist in state-space. It is the job of the investigator to determine what drives variation in the two rates. If juvenile density dependence is expected to be important driver, then researchers will need to measure growth and survival rates at relevant densities. Because current densities of FAS are extremely low compared to what the environment can support when predators are excluded (Barrus et al. 2023), we did not consider negative density dependence at the juvenile stage to be a good explanation for the demographic rates we observed in the field.

~~The negative slope to the isocline of juvenile growth and survival rates was a qualitative prediction from our model evaluation and is the expected result from extensive empirical studies that have explored how growth mediates mortality rates from consumers through a sensitive stage(Jeyasingh and Weider 2005, Craig et al. 2006, Davidson and Dorn 2018, Pepi et al. 2018,~~

~~Ma et al. 2021, Meehan et al. 2022, Davidson et al., 2024). The negative slope of the isocline result was consistent for different of hydrologic conditions that mediate reproductive output of the adult snails (Figure 2), including hydrologic conditions that were held constant at the ideal depth for reproduction (Appendix S1: Figure S3). Additionally, we included three examples which growth and survival rates were sufficiently studied in the field to highlight their potential for applying this approach (Figure 1). Furthermore, studies of the theoretical interactions between juvenile growth and survival scaling to population growth remain rare; our literature search found only one theoretical study exploring fish recruitment (Rice et al. 1993), and one theoretical study exploring how equilibrium densities of predator and prey populations are affected by juvenile growth and survival (Pepi et al. 2023). Although shown in a specific case, we expect the negative slope of the isocline would hold for any species with size-dependent survival driven by consumers.~~

The negative slope ~~to~~of the FAS isocline ~~of juvenile growth and survival rates~~ was a qualitative prediction from our model evaluation and is the expected result from extensive empirical studies that have explored how growth mediates mortality rates from consumers as juveniles pass through a sensitive- stage (Jeyasingh and Weider 2005, Craig et al. 2006, Davidson and Dorn 2018, Pepi et al. 2018, Ma et al. 2021, Meehan et al. 2022, Davidson et al., 2024). The negative slope of the isocline result was consistent for different ~~of~~ hydrologic conditions that mediate reproductive output of the adult snails (Figure 2), including hydrologic conditions that were held constant at the ideal depth for reproduction (Appendix S1: Figure S3). Additionally, we included three examples -which growth and survival rates were sufficiently studied in the field to highlight their potential for applying this approach (Figure 1). ~~Furthermore,~~ ~~s~~Studies of the theoretical interactions between juvenile growth and survival scaling to population growth remain rare; our literature search found only one theoretical study exploring fish recruitment (Rice et al. 1993), and one theoretical study exploring how equilibrium densities of predator and prey populations are affected by juvenile growth and survival (Pepi et al. 2023). Although shown in a specific case, we expect the negative slope of the isocline would hold for any species with populations experiencing high recruitment variation that is driven by consumers.~~with size-dependent juvenile survival driven by consumers.~~

~~Adjusting conditions for better FAS reproduction steepened the slope and effectively increased the combinatorial parameter space that produced increasing populations. This suggests that populations would disproportionately benefit from increased reproductive rates when juvenile growth rates were faster than when they were slower. The steeper slope was probably caused by the simulated juvenile snails hatching earlier and growing to maturity before the end of the summer reproductive season. To the best of our knowledge, this is possible under the current understanding of FAS biology, but has not been demonstrated, thus indicating an emergent hypothesis for this system that could be tested further. In the FAS population model, water depth was an environmental condition that influenced reproductive rates, but other environmental conditions that influence reproduction might produce similar results. Indeed, temperature effects on reproductive rates have been extensively studied for many organism(Cattau et al. 2014)s (Cattau et al. 2014, Dougherty et al. 2024), and temperature has been shown to influence multi-voltinism in moths in Boreal systems (Pöyry et al. 2011). More work is needed to corroborate if populations disproportionately benefit from increased reproductive rates in times or places with higher juvenile growth, but our work demonstrates a~~

~~need to explore the population dynamic outcomes of size- and growth-mediated juvenile mortality.~~

When measuring survival and growth in the field, the primary interest is where the parameters fall ~~on the~~relative to the isocline. And making specific predictions about a species of conservation concern *requires* a specific model for that species. Demographic models are becoming a widespread tool for conservation and ecology (Gacoigne et al. 2023), to the best of our knowledge, our work is the first to ~~not only~~ demonstrate their use in interpreting survival and growth rates measured in natural environments. E~~E~~mploying demographic~~ based~~ isoclines in population research would require the development of a size-indexed demographic model that allows the isolation of the growth and survival of sensitive juvenile stages. The time steps in the model ~~should be~~should be short relative~~ relevant~~ to the length of the sensitive stage~~s (size-dependent mortality)~~ and the measurements in the field should be measured commensurately to match the theoretical predictions.

Interpreting empirical measures of survival and growth

Including an isocline analysis of survival and growth allowed us to interpret natural empirically-measured parameters in a population dynamic perspective and offers insights about how environmental variation might influence consumer-resource interaction strength (Davidson et al. 2021, Pepi et al. 2023). Recent studies of temperature-~~dependenet~~dependent rates~~e~~ concluded that consumer-resource interaction strength should weaken or strengthen depending on asymmetries between thermal responses of the resource growth rate and consumer per-capita foraging (Davidson et al. 2021, Pepi et al. 2023). For our wetlands, the per capita foraging rates of ectothermic predators increased in the warmer wet season (calculation in Appendix S2: Figure S4), which should strengthen interactions between FAS and their predators (Figure 2) except that

~~lower~~ predator abundances after the wetlands reflooded ~~also changed~~ decreased between seasons (Appendix S2: Figure S4). The reduction in predator ~~community (~~abundances~~)~~ ~~changes~~ appears to have overwhelmed any increase in ~~changes~~ in ~~snail survival that~~ per capita consumption~~were~~ that was mediated by warmer temperatures ~~on per-capita foraging~~ (Appendix S2). Studies that isolate the effects of variable environmental conditions on predator-prey interactions have typically controlled predator densities experimentally, or statistically (Jeyasingh and Weider 2005, Davidson and Dorn 2018, Pepi et al. 2018, Davidson et al. 2021, Ma et al. 2021) to measure per-capita rates. But the study of natural populations in communities requires consideration of natural seasonal variation in predator abundances that could covary with temperature (Davidson et al. 2024). Environmentally-~~-~~mediated changes in predator communities may be more important to survival than the per-capita foraging rates of the predators and could ~~conceivably~~ either counteract or exacerbate the temperature-mediated changes to per-capita foraging rates.

Novel Hypotheses for Population Management

The hydrologic conditions within the Everglades are heavily managed with the goal of restoring conditions for wildlife and fishes, historic habitat mosaic~~s~~, and biodiversity, while~~, and~~ safe-guarding drinking water (National Academies of Sciences, Engineering, and Medicine 2021). Improving conditions for the FAS populations in the Everglades will be necessary to delist the federally endangered Everglades Snail Kite. The current paradigm for encouraging population growth of the FAS is to make hydrologic conditions more favorable for reproduction (i.e., maintaining water levels in a shallow range of depths that encourages egg laying; Darby et al. 2015), but our results indicate that with the current levels of predation and individual growth, improving hydrologic conditions for reproduction in the Everglades can, at best, only maintain the already small populations of FAS. For FAS population growth to become positive, we offer three hypotheses (~~see numbers in~~ Table 1~~ and Figure 3~~) about the spatiotemporal environmental conditions that could shift the average daily survival and growth conditions experienced by juveniles.~~ (open circles Figure 3).~~

The dry season parameters were ~~combinatorially~~ worse than the wet season parameters which is particularly problematic because most egg-laying occurs during the dry season (spring) before the water reaches its annual minimum depth (typically in May; Barrus et al. 2023). This result suggests that FAS may benefit more by improving dry season conditions for survival and growth than by improving wet season conditions. ~~Also, if~~If females can store their resources and hydrologic conditions ~~can~~ could conceivably shift more of the egg laying to the wet season, then the average demographic parameters would move up and right towards stasis or growth (Hypothesis 1-Table 1). Although more research is needed to understand how water levels might mediate this response, one observation suggests that shifting reproduction to the wet season (July–August) can occur in shorter-hydroperiod wetlands outside the ridge-slough landscape (O'Hare 2010).

Improved food quality could also move parameters to the right in state space (Hypothesis 2-Table 1). The Everglades is phosphorus-limited with natural oligotrophic total phosphorus concentrations between 110–400 µg·g^{-1} in the ridge and slough landscape (Gaiser et al. 2011). Growth of juvenile FAS varies with TP concentrations in the periphyton (Hansen et al. 2022, Barrus et al. 2023), and previous experimental manipulations of phosphorus showed that higher TP increased growth and juvenile apple snail survival in the presence of ~~gape-limited~~ crayfish (Davidson and Dorn 2018). Our results build on this finding by indicating that TP can mediate the net community level effects of predators on recruitment in the field. Periphyton TP levels were highest at WCA3A site 2 (Table S3), and it was the only site that had wet season growth and survival parameter combinations that predicted population growth. Nevertheless, restoration and management actions ~~expressly~~ attempt to avoid eutrophication of the Everglades (National Academies of Sciences, Engineering, and Medicine 2021). Perhaps more promisingly, the pre-drainage Everglades was a flowing ecosystem (the "River of Grass") with water velocities > 2 cm/s. Flow velocity was shown to increase growth of *Pomacea* apple snails through changes to microbial food quality (Hansen et al. 2022), and t~~T~~herefore~~, flow restoration~~ restoration of flowing water might improve growth rates of juvenile FAS (Hypothesis 2-Table 1).

Finally, the predation rates in the Everglades might currently be higher than historical levels as a function of non-native fishes (Pintar et al. 2023) or hydrologic conditions supporting higher densities of juvenile invertebrate predators ~~(e.g., invertebrates)~~ in the sloughs (Hypothesis 3-Table 1). Some non-native fishes introduced to the Everglades have molluscivorous tendencies, but our observations of diets and tethering ~~remnants~~ remains suggested that native predators (e.g., crayfish, giant water bugs, greater sirens) in LILA seem to be more responsible for mortality patterns than non-native species like Mayan cichlids (*Mayaheros uropthalmus*). Because the predator assemblage feeding on juvenile snails included native species existing across a wide range of the hydroperiod gradient, it remains unclear how ~~floods~~ water depths or hydrologic droughts would shift survival rates (e.g., drying benefits crayfish populations; Dorn and Cook 2015, Sinnickson and Dorn 2024).

Conclusion

We demonstrated the first demographic isocline approach ~~to studying spatiotemporal recruitment variation in populations~~ that could be used when studying ~~in studying~~ and managing size-structured consumer-resource interactions ~~across terrestrial, marine and freshwater ecosystems~~.

The demographic isocline was conceptually derived from well-studied principles of size-structured interactions (Werner and Gilliam 1984, Davidson et al. 2024), and the operationalized isocline was built using a size-indexed demographic model and ~~from~~ theoretical combinations of daily survival and growth. ~~of sensitive sizes within a size-indexed demographic model and~~ The isocline illustrated ~~a~~ the expected negative relationship between juvenile growth and survival~~al.~~ ~~The negative relationship indicates that;~~ populations with faster-growing juveniles can withstand (λ=1) greater mortality (lower survival) ~~and still grow~~. A demographic ~~When this approach is operationalized in a field setting, the~~ isocline can be used as a map to interpret empirically measured rates in a ~~local~~ population dynamic contexts ~~can be interpreted from field-based demographic rates~~. Our case study indicated that seasonal changes in the predator community were more important in determining interaction strength than simpler physiological expectations based on thermal responses of predators and prey. This demographic isocline approach provided novel hypotheses about the conditions needed to restore a historical resource of an endangered species. We encourage others to consider further theoretical exploration of the concept and development of size-indexed models and demographic isoclines to study populations with high recruitment variability.

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Author Contributions

All authors contributed to the design. The tethering and growth rate design and experiments were established by NJD, NTB and MIC. Development of the isocline was conducted by NTB in consultation with NJD. Data collection was performed by NTB and NJD. Analyses and statistical models were conducted by NTB in consultation with NJD. The paper was written by NTB and NJD with edits from MIC. All authors have read and approved the final manuscript.

Conflict of Interests

We declare no financial interests that could create conflicts for this work.

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Tables

Table 1: Three hypothesized changes in spatiotemporal conditions that could shift juvenile FAS survival and growth parameters from the left (declining) side of the isocline to the right side of the isocline.

Hypothesis	Predicted Parameter Shift	Description of Environmental Change Hypothesis
1	Up and right	Seasonal depth/hydro-patterns that shift more of the egg production to the wet season expose more juveniles to favorable wet season parameters.
2	Right	Hydrologic conditions that improve growth rates (e.g., flow or flow-loading of nutrients) through improved food quality or water quality (oxygenation).
3	Up	Hydrologic patterns that disfavor important predators could improve survival for juvenile snails (especially in the dry season).

Figure Captions

Figure 1. A) Zero-population growth isocline illustrating the expected joint impact of growth rates and mortality to consumers at a sensitive stage (diagonal black line). Resource populations are decreasing/recruiting in areas to the left and below the isocline shaded purple) while areas above and to the right indicate populations that are increasing/recruiting (shaded green). The strength of consumer control is represented by the gradient from dark green to dark purple. B-D) Examples of populations spanning freshwater, terrestrial and marine ecosystems for which hypothetical demographic isoclines could provide conceptual meaning to population dynamics and predictive values for field measured rates. The color of points corresponds to their position along the gradient in A). B) The relationship between mosquito (*Aedes atropalpus*) population with predatory dragonflies (*Pantala* spp.) depends on joint temperature mediation of survival and growth of mosquitos. C) Quaking aspen (*Populus tremuloides*) stand recruitment depends on moisture/precipitation gradients and elk (*Cervus canadensis*) browsing rates driving sucker survival. D) Oyster (*Crassostrea virginica*) growth and mortality to predatory drilling snails are affected by salinity gradients, though growth rate declines as seen from low to intermediate level have not been described at high salinity levels and could follow linear or unimodal relationships. All vector graphics except the dragonfly and mosquito larvae are from the University of Maryland's Integration and Application Network (ian.umces.edu/media-library). The dragonfly and mosquito larvae are from Aquatic Ecology Lab at Florida International University (FIU), principal investigator Nathan J. Dorn.

Figure 2 Isoclines illustrating the ~~combinatorial~~ interactive effects of juvenile growth and survival that produce zero net population growth for a size-structured model of a freshwater gastropod (*Pomacea paludosa*) under two different hydrologic regimes that affect reproduction (black isocline = lower reproduction, gray isocline = higher reproduction). Points are mean daily survival (snails < 10mm SL) and growth (k_{growth}) quantified in LILA wetlands and site 2 in Water Conservation Area 3A. Error bars represent 95% confidence intervals for each parameter estimate. The combined parameters (open symbols) were calculated by a weighted average reflecting greater juvenile snail production (egg laying and hatching) in the dry season.